INTERNATIONAL JOURNAL OF REMOTE SENSING
2024, VOL. 45, NO. 9, 3075–3103
https://doi.org/10.1080/01431161.2024.2343132

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A novel graph-based multiple kernel learning framework for hyperspectral image classification

Shirin Hassanzadeh[a], Habibollah Danyali[a], Azam Karami[b] and Mohammad Sadegh Helfroush[a]

[a]Department of Electrical Engineering, Shiraz University of Technology, Shiraz, Iran; [b]Department of Physics, Shahid Bahonar University of Kerman, Kerman, Iran

ABSTRACT

Multiple kernel learning (MKL) is an efficient way to improve hyperspectral image classification with few training samples by integrating spectral and spatial features. Nonetheless, presenting a MKL algorithm to capture significant information and eliminate redundancy is indispensable. In this paper, a novel graph-based MKL method (GMKL) is introduced to enhance the performance of the MKL algorithm and achieve high discriminative features, resulting in maximum separability. To do so, an informative low-rank graph is determined via a new algorithm called block diagonal representation that takes into account both local and global data structures. Then, in lieu of a time-consuming search for finding the optimal combination of the basic kernels, unlike the prior works, an optimal projective direction is acquired using graph theory. Various basic kernels generated from superpixels and attribute profiles are projected to produce a discriminative combination of kernels. Eventually, the optimum kernel is given to the support vector machine classifier. Experiments have been conducted on two popular hyperspectral datasets (Indian Pines and Pavia University). For the datasets, the proposed algorithm gained 90.43% and 99.28% overall accuracy, respectively, using only 1% training samples, which is much better than the state-of-the-art MKL algorithm and deep learning-based approaches with lower computational time.

ARTICLE HISTORY
Received 8 October 2023
Accepted 26 March 2024

KEYWORDS
Classification; hyperspectral data; image processing

1. Introduction

Hyperspectral images (HSI) have emerged as one of the most commonly used remote sensing images in a wide range of applications, including precision agriculture (Y. Liu, Pu, and Sun 2017; Xing et al. 2019), mine exploration (Götze, Beyer, and Gläßer 2016; Sven, Murphy, and Melkumyan 2014), environmental monitoring (Honkavaara et al. 2014) and military applications (Briottet et al. 2006). The most widely used method for analysing hyperspectral images is classification, which is utilized in a variety of applications, for instance, accurate extraction and mapping of floating raft aquaculture (FRA) (Hou et al. 2022) and vegetation and wetland mapping (Huang et al. 2023; G. Yang et al. 2022).

CONTACT Habibollah Danyali danyali@sutech.ac.ir Department of Electrical Engineering, Shiraz University of Technology, Shiraz, Iran; Shirin Hassanzadeh s.hassanzadeh@sutech.ac.ir

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Owing to the high dimensionality of HSI, reducing the spectral dimension of the data is necessary, particularly if there are few training samples (Hughes phenomenon) (Bioucas-Dias et al. 2013). To reduce the dimension, some feature extraction approaches based on spectral information were applied to classify HSI (W. Li et al. 2018). Principal component analysis (PCA) (Licciardi et al. 2012) and linear discriminant analysis (LDA) (Bandos, Bruzzone, and Camps-Valls 2009) are two common feature extraction techniques that map high-dimensional spectral features into low-dimensional ones without taking into account the intrinsic nonlinear structure of HSI. There are other dimension reduction techniques based on graph structure that consider nonlinearity characteristics (Hassanzadeh, Danyali, and Sadegh Helfroush 2022a). In graph-based methods (L. Zhang and Luo 2020), an undirected weighted graph has been constructed to exploit the geometrical properties of the original data, and a projection matrix has been estimated by solving a generalized eigenvalue problem. Band selection is another strategy to reduce HSI dimension that maintains the most informative bands and alleviates the computational cost. Many approaches have been suggested for band selection; in the most recent research, a subset of bands was selected by a band evaluation index obtained using a dual global-local attention network (He et al. 2022).

Considering the above, some classifiers utilize spectral features as input; namely, support vector machine (SVM) is a powerful supervised classification method for HSI (Sheykhmousa et al. 2020). Sparse representation classification (SRC) (W. Li and Du 2016) is another spectral-based classifier in which each pixel has been estimated using a sparse combination of all samples (dictionary). The SRC is time-consuming and requires a large number of training samples to achieve good performance. When spatial information is incorporated with spectral information, the HSI classification accuracy is significantly enhanced. Therefore, some spatial-spectral classifiers that are based on both spectral and spatial information have been developed. In (Jiang et al. 2018), to improve the performance of the SVM classifier, PCA has been applied to each homogeneous region obtained by super-pixel segmentation. Additionally, a stacked vector of multi-scale spatial features, as Gaussian pyramids demonstrated in (Tu et al. 2019), increases classification accuracy. Another feature extraction method based on tensor decomposition considers both spectral and spatial information (Deng et al. 2018), so that the intrinsic geometrical structure of the input data is preserved. To capture a variety of region-based attributes, invariant attribute profiles (IAP) have been applied in both spatial and Fourier frequency domain (Hong et al. 2020). Then, different IAPs were stacked as a vector and given to a random forest or nearest neighbour classifier (Bo, Lu, and Wang 2017; Cao et al. 2019).

Deep neural network-based classifiers have received considerable attention during the past few years (H.-C. Li et al. 2023). Some neural network-based methods, such as auto-encoders, have been proposed to learn deep features directly from the spectral information and classify hyperspectral images efficiently based on the features (Ma, Wang, and Geng 2016). Following the development of the convolutional neural network (CNN), different CNN-based structures for HSI classification were investigated. These techniques have a large number of parameters to be trained, and the number of training samples should be sufficiently large to achieve high performance. The performance of CNN was improved by introducing a pixel-pair model (W. Li et al. 2017) in which each central pixel of the test data has been combined with each of its surrounding pixels, and the final label has been assigned using majority voting.

Recurrent neural networks (RNN) (Hang et al. 2019) were designed to regard the HSIs as sequential data, and the relationship between spectral bands was captured via recurrent operators. A deeper and wider NN-based method was suggested (J. Yang et al. 2021) with two subnetworks. The upper network extracted multiscale spectral features based on cascaded dilated convolutional layers, and the lower one consisted of multiple parallel paths for multiscale spatial features. A different multi-branch network was developed to acquire robust and more precise information through the fusion of multi-scale spectral and spatial features (Yang et al. 2023). To achieve a better and more stable local optimum solution in the training phase, a graph-based convolutional neural network (GCN) (Hong et al. 2021) has been presented in minibatch mode. These miniGCNs can infer the new input samples and obtain better classification performance without retraining networks, as well as reduce the huge complexity of GCN. A dual-branch residual spectral-spatial network (Ghaderizadeh et al. 2022) has gained classification accuracy by effectively extracting multiscale spectral and spatial features based on the CNN, then applying an attention mechanism to emphasize significant features, and eliminating redundant features. In addition to CNNs and RNNs, transformer-based methods have also been suggested for improving deep learning-based classifiers (J. Yang, Du, and Zhang 2023). A CNN-based method with tokenization transformer has been introduced (L. Sun et al. 2022) to model low-level spectral and spatial features and capture high-level semantic ones that make a better trade-off between accuracy and computational cost. Also, spectralformer techniques (Y. Sun et al. 2023) have been introduced to fully utilize the discriminative advantage of the HSI spectrum, which enhances CNN performance by more effectively leveraging spectral data. To consider coarse-to-fine spatial information, the spatial-logical aggregation network (SLA-NET) is an efficient way to integrate spatial convolutional and morphological operations into a single architecture and reduce information loss during data transmission in the deep network (M. Zhang et al. 2023).

Some shortcomings of deep learning-based methods have been alleviated by a full model considering three components that include extensive exploration of available features, reuse of representative features and multidomain feature fusion (J. Yang, Bo, and Liangpei 2023). And although employing rich spatial and spectral information can enhance classification accuracy, most of these algorithms have lower performance with limited training samples. Some recently developed deep learning models, such as few-shot learning (Q. Liu et al. 2023) and domain adaptation (Huang et al. 2022), have led to improved accuracy of HSI classification by utilizing abundant unlabelled data. However, both methods require a large number of labelled data in the pretraining stage. Self-supervised learning (SSL) (Liu, K. Liu et al. 2023) is another novel HSI classification in the field of unsupervised learning that includes two phases: self-supervised pretraining to extract deep features based on pretext tasks and fine-tuning for better feature learning. These models, despite their superior classification accuracy, are complex and have a number of parameters that necessitate learning.

Multiple kernel learning (MKL)-based classifiers have been presented to address the issue of low accuracy in a few training samples. Using MKL, different spatial and spectral features are integrated optimally, and significant features leading to high class discrimination can be preserved (Hassanzadeh, Danyali, and Sadegh Helfroush 2022b). In (Gu et al. 2012) PCA was employed for kernel learning and to obtain the optimal kernel combination. First, several base kernels with different

Gaussian scales were generated, and kernel weights were then calculated using the eigenvectors, which correspond to the maximum variance component. Linear discriminant analysis (LDA) is another subspace learning method to compute the weights of base kernels (Wang, Gu, and Tuia 2016) and outperforms PCA in terms of classification accuracy and computational cost. To elevate the classification performance of MKL-based methods, some feature extraction techniques based on tensor model and morphological profiles were suggested (Gu, Liu, and Li 2019; L. Sun et al. 2019) that fully utilize the spatial texture of HSI. Incorporating multiscale superpixels and multi-attribute profiles (EMAPs) features (Pan et al. 2020) has substantially enhanced the performance of PCA-based MKL. Although an ensemble MKL method based on boosting strategies (Gu et al. 2017) can achieve better accuracy, it is computationally costly. To reduce the computational complexity, a MKL-based classification algorithm has been presented (T. Liu et al. 2016), in which only the elements of kernel matrices corresponding to each of the two classes were included in the learning procedure. Furthermore, the classification outcomes of the optimal kernel combination based on the Hadamard product among base kernels (Gu et al. 2016) revealed that the nonlinear MKL algorithm has lower performance than the linear one. A new deep MKL method based on one-dimensional CNNs (Ansari et al. 2022) was introduced to estimate a low-rank approximation of the basis kernels. Although these methods lead to better performance at low training rates, they do not achieve high discriminative power while reducing the redundancy of kernels.

In this paper, we propose a novel multiple kernel learning algorithm for HSI classification using a graph-based discriminative analysis. The main objective of the kernel learning process is to identify key kernels that result in the highest class separation while minimizing redundancy between kernels. To construct some base kernels, in the proposed method, first, various feature sets based on attribute profiles and different scales of superpixel are created. The superpixels are extracted based on a real-time segmentation algorithm to reduce the computation time. Then, a graph-based learning method is applied to the base kernels to find a projection direction. All basic kernels are projected to generate an optimal combined kernel. Finally, the obtained kernel is substituted in the support vector machine (SVM) classifier. The proposed graph-based MKL (GMKL) boosts kernel discriminative power and thus classification accuracy, especially when only a few training samples are available. This is primarily due to optimally incorporating diverse features into the classification process while eliminating redundant information. The main contributions of this paper are as follows:

- A graph-based kernel learning model is introduced to exploit the intrinsic spatial or spectral relationships of the hyperspectral image and determine the optimum combined kernel.
- A block diagonal representation algorithm based on subspace learning is proposed to obtain the graph matrix in order to maximize class separability.
- A real-time superpixel-based feature extraction is adopted to decrease the computational cost.

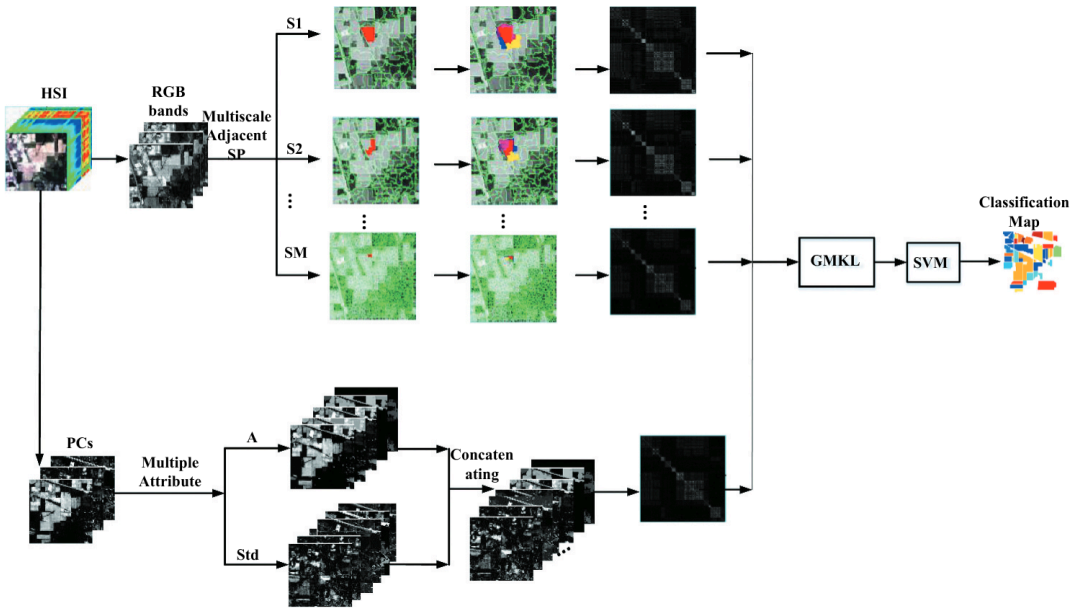


Figure 1. Block diagram of the proposed GMKL method.

2. Proposed method

The proposed method, GMKL, is depicted in Figure 1. As shown in the figure, different kernels are built based on multiscale superpixels and attribute profile features, and the optimized combination of kernels is then embedded in the SVM classifier. The main steps of the proposed GMKL are as follows:

(1) First, three principal components and RGB bands are obtained from the original HSI.
(2) Feature sets are generated based on morphological attribute profiles and multiscale adjacent superpixels from PCs and RGB bands, respectively.
(3) Multiple Gaussian kernels are computed from the feature sets.
(4) The kernel learning procedure is performed based on graph discriminant analysis.

2.1. Adjacent superpixel extraction

Superpixel segmentation techniques are widely used in computer vision and image processing. Each superpixel is a homogeneous region whose form and size may be changed adaptively based on various image structures (Tu et al. 2020).

In the proposed method, to decrease the computing cost of superpixel segmentation, a real-time density-based spatial clustering of applications with noise (DBSCAN) (Shen et al. 2016) has been employed. This method overcomes the drawback of the original DBSCAN clustering algorithm, which is sensitive to parameters, in order to generate satisfactory clustering results. The basic idea behind the DBSCAN clustering algorithm is that the density of points within each cluster is significantly higher than it is outside. Also, the density of noisy areas is lower than

that of each cluster. DBSCAN has remarkable potential for handling noise and outliers, as well as for segmenting complex and irregularly shaped objects. On the other hand, it has shown high performance in segmenting low-contrast images due to its strong edge adherence (L. Zhang et al. 2021). Therefore, it is both more accurate and efficient than previous superpixel algorithms. Real-time DBSCAN is a two-stage clustering and merging algorithm based on three spectral bands (RGB) that improves the performance of segmentation techniques by adding geometric and colour constraints. First, this algorithm creates initial superpixels based on four neighbouring pixels of labelled pixels using a search algorithm, then refines these initial superpixels by merging very small ones to produce the final superpixel results; for more details see (Shen et al. 2016).

During the merging phase, all the initial superpixels $I(p)$ are processed from left to right and from top to bottom. If the number of pixels in one initial superpixel is less than a specific threshold, another initial superpixel from its nearest neighbour will be merged into this one. After the merging phase, the algorithm obtains the final refined superpixels with the regular shapes. These two steps are formulated as Equation 1 and Equation 2.

$$I(p) = \{i \in I | D_1^p(i,j) < \varepsilon, i \sim j, j \in p\} \cup I(p) \tag{1}$$

where p is a superpixel, ε is a threshold that controls the number of superpixels the user requires, it is calculated by dividing the size of the input image by the total number of superpixels. $i \sim j$ indicates pixel i around pixel j, j is in the labelled set $I(p)$ and $D_1^p(i,j)$ is the distance between pixels i and j. The initial superpixels are yielded during the clustering phase using Equation 1.

$$\begin{aligned} I(i) &= \min D_2(p,i), i \sim p \\ I_R(p) &= I(p) \cup I(i) \end{aligned} \tag{2}$$

Equation 2 is used in the merging phase, in which $D_2(p,i)$ is the distance between initial superpixels p and i.

Calculating the distance metric is an important step in the DBSCAN algorithm, which is defined in the clustering phase as a linear combination function$D_1(i,j)$:

$$d_{adj}(i,j) = \sqrt{(R_i - R_j)^2 + (G_i - G_j)^2 + (B_i - B_j)^2}$$

$$d_{set}(i,k) = \sqrt{(R_i - R_k)^2 + (G_i - G_k)^2 + (B_i - B_k)^2}$$

$$D_1(i,j) = c_1 d_{adj}(i,j) + c_2 d_{set}(i,k) \tag{3}$$

where $d_{adj}(i,j)$ is the distance between two neighbouring pixels i, j in RGB colour space and $d_{set}(i,k)$ is the distance between an unlabelled pixel i and a labelled pixel k (the first top-left pixel). R_i, G_i and B_i denote the RGB colour features of pixel i; c_1, c_2 are the weight parameter.

In the merging phase, the distance function is computed between the initial superpixels by a linear combination of colour distance and spatial distance:

$$d_c(I,p) = \sqrt{(\bar{R}_I - \bar{R}_p)^2 + (\bar{G}_I - \bar{G}_p)^2 + (\bar{B}_I - \bar{B}_p)^2}$$

$$d_a(l,p) = \sqrt{(\bar{x}_l - \bar{x}_p)^2 + (\bar{y}_l - \bar{y}_p)^2}$$

$$D_2(l,p) = d_c(l,p) + c_3 d_a(l,p) \tag{4}$$

where $d_c(l,p)$ is the colour distance between two initial superpixels l and p and shows a degree of similarity; $d_a(l,p)$ is the spatial distance between two pixels, which has a significant impact on the shape of the final superpixels. $\bar{R}_p$, $\bar{G}_p$ and $\bar{B}_p$ are the mean of RGB colour values for initial superpixel p, $\bar{x}_p$ and $\bar{y}_p$ are the coordinates of the centroid in superpixel p, c_3 is a positive balancing parameter. In our proposed method, to achieve a balance between accuracy and computational cost, parameters $c_1 = 0.6, c_2 = 0.4, c_3 = 1$ were set, and ε should be calculated for each dataset and each number of superpixels, separately.

After obtaining superpixels by DBSCAN, HSI $H \in \mathbb{R}^{\mathbf{B}\times\mathbf{N}}$is divided into superpixels, $\{\Omega_1, \ldots, \Omega_i, \ldots, \Omega_T\} \in \mathbb{R}^{B\times N}$, where T is the total number of superpixels, $N = \sum_i N_i$ is the total number of pixels in each spectral band, and N_i is the number of pixels in each superpixel. In the proposed method, base kernels are constructed from adjacent superpixel (AS) strategy. The adjacent superpixel set of a single superpixel $\mathbf{\Omega}_i$ is $\left\{\mathbf{\Omega}_{i1}, \ldots, \mathbf{\Omega_{ik}}, \ldots, \mathbf{\Omega}_{\widehat{\boldsymbol{im}}}\right\}$, where is the number of superpixels adjacent to a target superpixel in a homogeneous region. Consider each $\phi(\mathbf{\Omega}_{ik})$ to be a subcluster in kernel space; merge these subclusters into a single cluster ($\mathbf{\Omega}_{ik}^{mean}$) and assign different weights to each subcluster to reduce the effects of undesirable pixels. The kernel element corresponding to *Accessisdenied* and $\mathbf{x}_j \in \mathbf{\Omega}_j$ is determined as follows:

$$\begin{aligned}\mathbf{K}^{\mathrm{AS}}(\mathbf{x}_i, \mathbf{x}_j) &= \sum_{k=1}^{\hat{m}} \sum_{l=1}^{\hat{n}} \langle \phi(w_{ik}, \mathbf{\Omega}_{ik}^{mean}), \phi(w_{jl}, \mathbf{\Omega}_{jl}^{mean}) \rangle \\ &= \frac{1}{\sum_k \sum_l w_{ik} w_{jl}} \sum_{k=1}^{\hat{m}} \sum_{l=1}^{\hat{n}} w_{ik} w_{jl} \mathbf{K}(\mathbf{\Omega}_{ik}^{mean}, \mathbf{\Omega}_{jl}^{mean})\end{aligned} \tag{5}$$

where $\mathbf{\Omega}_{ik}^{mean}$ is the mean value of $\mathbf{\Omega}_{ik}$, and and $\hat{n}$ are the adjacent superpixel numbers of $\mathbf{\Omega}_i$ and $\mathbf{\Omega}_j$, respectively. The correlation of centre superpixel and its neighbours is shown by $w_{ik} = \exp(-2r_0 \left\| \mathbf{\Omega}_i^{mean} - \mathbf{\Omega}_{ik}^{mean} \right\|_2^2)$, where r_0 is a degree of the filter.

2.2. Extended Attribute Profiles (EAP)

Morphological attribute profiles (APs) are an extension of morphological profiles (MPs) and are flexible tools to model different kinds of structural information using different types of attributes (Mura et al. 2010). Based on the type of attributes considered in the transformation, different parametric features can be modelled. In GMKL, two attributes are employed to extract spatial information: area (which is related to the MP created with a squared SE) and standard deviation (which is a descriptor of the geometry of a region invariant to the scale). When compared to traditional MPs, the computational load required to create APs has been reduced. In addition, by concatenating multiple APs, extended morphological attribute profiles (EAPs) are formed. EAPs improve spatial information modelling and offer a more

accurate description of the scene (Fang et al. 2018). Finally, the optimum combination of kernels is learned from multiscale superpixels and EAPs, using the GMKL algorithm.

2.3. Graph-based Multiple Kernel Learning (GMKL)

Suppose $\{\mathbf{x}_i\}, i = 1, 2, \ldots, N$ is a set of labelled samples and $\{y_i\}$ are labels, SVM problem for the samples is defined as a maximization problem:

$$\max_{\alpha}\left\{\sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i,j}^{n}\alpha_i\alpha_j y_i y_j \mathbf{K}(\mathbf{x}_i, \mathbf{x}_j)\right\} \tag{6}$$

where α_i is the classification parameter and $\mathbf{K}(\mathbf{x}_i, \mathbf{x}_j) = \Phi(\mathbf{x}_i), \Phi(\mathbf{x}_j)$ is the kernel matrix with a mapping function $\Phi(\mathbf{x})$. The kernel function can be a linear, polynomial or radial basis function (RBF). An MKL algorithm is a weighting process that is used to determine which key kernels should be kept and how important they are in the optimal combination of basic kernels. Most MKL algorithms have used RBF (Gaussian) kernel since it is more efficient in terms of classification accuracy and the most representative kernel function with certain properties such as translation invariability. The Gaussian kernel is defined by:

$$exp\left(\frac{||x_i - x_j||^2}{2\sigma^2}\right), \sigma \in \mathbb{R}^+ \tag{7}$$

where σ is the scale (variance) of the kernel. The kernel's scale plays an important role in classification, and it is preferable to choose this parameter for each dataset individually.

To incorporate different spatial and spectral features and also various modalities of original data, linear or nonlinear MKL methods have been presented. Because linear techniques appear to be more acceptable when fusing complex Gaussian kernels (Gu et al. 2016), we consider a linear MKL problem that is defined as:

$$\min_{\eta_m}\max_{\alpha}\left\{\sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i,j}^{n}\alpha_i\alpha_j y_i y_j \sum_{i=1}^{m}\eta_m \mathbf{K}(\mathbf{x}_i, \mathbf{x}_j)\right\} \tag{8}$$

subject to $0 \le \alpha \le C, \sum_{i=1}^{N}\alpha_i y_i = 0, i = 1, 2, \ldots, N$ and $\sum_{m=}^{M}\eta_m = 1, \eta_m \ge 0$

where M is the total number of base kernels, and η_m is their weights in the model. The weights are calculated in the kernel learning procedure by fixing α_i, and then this parameter is optimized by fixing the weights. As a result, the goal is to find a kernel learning algorithm that produces the highest class separation and feature discrimination. Although various kernel learning methods have been suggested, their performance in terms of both computational cost and accuracy is inefficient. In this paper, we propose a novel MKL technique with low computational complexity based on low-rank graph discriminant analysis, which leads to high accuracy and will be discussed in detail in the following section.

Graph-based discriminant analysis (GDA) has been proposed in this paper as a novel MKL algorithm to attain the optimal linear combination of base kernels. As mentioned, the kernel is a measure of similarity, and an ideal kernel is achieved by maximizing within-

class similarities, and minimizing between-class similarities. Thus, it is critical to combine base kernels in such a way that greater class separability is obtained and the optimal kernel is as close to the ideal kernel as possible.

In the MKL problem, the objective is to compute the weights of a linear combination of given basis matrices $\mathbf{K}_m$:

$$\begin{aligned}\mathbf{K}(\mathbf{x}_1,\mathbf{x}_2) &= \sum_{m=1}^{M}\eta_m\mathbf{K}_m(\mathbf{x}_i,\mathbf{x}_j)\\ s.t.\eta_m &\geq 0 \text{ and } \sum_{m=1}^{M}\eta_m = 1\end{aligned} \tag{9}$$

By concatenating the kernel matrices, a 3-D data cube is generated. To find a low-dimensional subspace of this cube, a graph-embedding subspace learning framework has been exploited. First, each kernel matrix is converted into a column vector of $\mathbf{Q}$ matrix $Q = [vec(K_1); vec(K_2); \ldots; vec(K_M)] \in \mathbb{R}^{M\times N^2}$ then an $M \times K$ projection matrix (with $K \ll M$) is found based on graph learning, which results in a low-dimensional subspace $\mathbf{Y} = \mathbf{P}^T\mathbf{Q}$, where desired statistical or geometrical characteristics are maintained (W. Li, Liu, and Du 2016). The objective function is defined as:

$$\mathbf{P} = \arg\min_{\mathbf{P}} \frac{\left|\mathbf{P}^T\mathbf{Q}\mathbf{L}\mathbf{Q}^T\mathbf{P}\right|}{\left|\mathbf{P}^T\mathbf{Q}\mathbf{L}_p\mathbf{Q}^T\mathbf{P}\right|} \tag{10}$$

where $\mathbf{L} = \mathbf{D} - \mathbf{W}, \mathbf{L}_p$ is the Laplacian matrix of graph $\mathbf{G}, \mathbf{D}$ is a diagonal matrix with $D_{ii} = \sum_{j=1}^{n} w_{ij}$, and $\mathbf{L}_p$ is the Laplacian matrix of the penalty graph $\mathbf{G}_p$.

Solving Equation 10 results in the following relationship, which is a generalized eigenvalue decomposition problem (Ghojogh, Karray, and Crowley 2019):

$$\mathbf{Q}\mathbf{L}\mathbf{Q}^T\mathbf{P} = \mathbf{\Lambda}\mathbf{Q}\mathbf{L}_p\mathbf{Q}^T\mathbf{P} \tag{11}$$

By considering $\sum = \mathbf{Q}\mathbf{L}\mathbf{Q}^{\boldsymbol{T}}$ and $\sum_p = \mathbf{Q}\mathbf{L}_{\boldsymbol{p}}\mathbf{Q}^{\boldsymbol{T}}$:

$$\Sigma\mathbf{P} = \mathbf{\Lambda}\Sigma_{\boldsymbol{p}}\mathbf{P} \tag{12}$$

where $\mathbf{P}$ and $\mathbf{\Lambda}$ matrices are calculated from the eigenvectors and eigenvalues of $\Sigma_{\boldsymbol{p}}^{-1}\Sigma$ matrix. The projection matrix $\mathbf{P}$ has dimensions of $M \times K$, where K is the number of eigenvectors, and $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues. The projection vector $\eta_m^* = [\eta_{11}^*, \eta_{12}^*, \ldots, \eta_{1M}^*]^T$, which is the eigenvector corresponding to the smallest non-zero eigenvalue, is then chosen as the optimal weight for the basis kernels. As a result, referring to Equation 9, the final optimal kernel is computed as follows from the basis kernels:

$$\mathbf{K}^* = \sum_{m=1}^{M}\eta_{1m}^*\mathbf{K}_m(\mathbf{x}_i,\mathbf{x}_j) \tag{13}$$

It is important to note that the performance of graph-based methods is primarily determined by the weights (similarity) matrix. In our proposed method, $\mathbf{K}_m, for\, m = 1, \ldots, M$ are obtained from different scales of superpixels and attribute profiles.

2.3.1. Block diagonal representation for GDA

To generate the weight matrix corresponding to the above Q matrix, a subspace learning problem based on the block diagonal regularizer is introduced in this paper to attain a block diagonal weight matrix $\mathbf{W}$ as:

$$\begin{aligned} &\min_{\mathbf{W}} \frac{1}{2}\|\mathbf{Q}-\mathbf{QW}\|^2 + \gamma\|\mathbf{W}\|_{\boxed{k}}, \\ &s.t.\ \mathrm{diag}(\mathbf{W}) = 0, \mathbf{W} \geq 0, \mathbf{W} = \mathbf{W}^T \end{aligned} \tag{14}$$

where $\mathbf{W}$ is a non-negative and a symmetric matrix, both of which are required conditions for construction of the block diagonal regularized matrix, $||\mathbf{W}||_{\boxed{k}}$ is k-block diagonal regularizer term and γ is a regularization parameter. For a matrix $\mathbf{W} \in \mathbb{R}^{n\times n}$, the k-block diagonal regularizer is determined as the sum of the k smallest eigenvalues (λ_i) of $\mathbf{L}_w$:

$$||\mathbf{W}||_{\boxed{k}} = \sum_{i=n-k+1}^{n} \lambda_i(\mathbf{L_W}) \tag{15}$$

where $\mathbf{L}_w = Diag(\mathbf{W}1) - \mathbf{W}1$ and $\mathbf{W}1 = \mathbf{W} \times ones(n, 1), n = size(\mathbf{Q}, 2)$.

The above minimization problem (Equation 14) is equivalent to:

$$\begin{aligned} \min_{\mathbf{Z},\mathbf{W}} \frac{1}{2}\|\mathbf{Q}-\mathbf{QZ}\|^2 + \frac{\lambda}{2}\|\mathbf{Z}-\mathbf{W}\|^2 + \gamma\|\mathbf{W}\|_{\boxed{k}} \\ s.t.\ \mathrm{diag}(\mathbf{W}) = 0, \mathbf{W} \geq 0, \mathbf{W} = \mathbf{W}^T \end{aligned} \tag{16}$$

$\mathbf{Z}$ is an auxiliary matrix to make the subproblems convex and λ is another regularization parameter. The non-convex term $||\mathbf{W}||_{\boxed{k}}$ can be reformulated as a convex problem (Lu et al. 2019):

$$\|\mathbf{W}\|_{\boxed{k}} = \min_{\mathbf{S}} \langle \mathbf{L_W}, \mathbf{S} \rangle, s.t.\ 0 \leq \mathbf{S} \leq \mathbf{I}, \mathrm{Tr}(\mathbf{S}) = k \tag{17}$$

So, Equation 17 is converted to:

$$\begin{aligned} &\min_{\mathbf{Z},\mathbf{S},\mathbf{W}} \frac{1}{2}\|\mathbf{Q}-\mathbf{QZ}\|^2 + \frac{\lambda}{2}\|\mathbf{Z}-\mathbf{W}\|^2 + \gamma\langle diag(\mathbf{W1}) - \mathbf{W}, \mathbf{S}\rangle \\ &s.t.\ \mathrm{diag}(\mathbf{W}) = 0, \mathbf{W} \geq 0, \mathbf{W} = \mathbf{W}^T, \\ &0 \leq \mathbf{S} \leq \mathbf{I}, \mathrm{Tr}(\mathbf{S}) = k \end{aligned} \tag{18}$$

Then, Equation 18 can be solved by alternating updating $\mathbf{S}, \mathbf{Z}$ and $\mathbf{W}$. Since the problem in Equation 18 with the k-block diagonal regularizer is nonconvex, it is difficult to optimize the problem using this kind of regularizer. The general Alternating Direction Method of Multipliers (ADMM) (Lu et al. 2018) is a potential solver to the related problem with a linear constraint and provides a convergence guarantee. This generalization of ADMM improves the efficiency of ADMMs and simplifies the optimization problem by replacing a non-separable objective function with a separable approximation function.

The following equations outline the optimization of $\mathbf{S}, \mathbf{Z}$ and $\mathbf{W}$. The ADMM sub-iterations for the proposed approach are also summarized in Algorithm 2. First, $\mathbf{S}, \mathbf{Z}$ matrices that have the same dimensions are updated by fixing $\mathbf{W} = \mathbf{W}^k$ as follows:

$$\begin{aligned}&\mathbf{S}^{k+1} = \arg\min_{\mathrm{S}} \langle diag(\mathbf{W1}) - \mathbf{W}, \mathbf{S}\rangle \\ &s.t.\, 0 \le \mathbf{S} \le \mathbf{I}, \mathrm{Tr}(\mathbf{S}) = k\end{aligned} \tag{19}$$

$$\mathbf{Z}^{k+1} = \arg\min_{\mathbf{Z}} \frac{1}{2}\|\mathbf{Q} - \mathbf{QZ}\|^2 + \frac{\lambda}{2}\|\mathbf{Z} - \mathbf{W}\|^2 \tag{20}$$

Second, fix $\mathbf{S} = \mathbf{S}^{k+1}, \mathbf{Z} = \mathbf{Z}^{k+1}$ and update $\mathbf{W}$ by:

$$\begin{aligned}&\mathbf{W}^{k+1} = \arg\min_{\mathbf{W}} \frac{\lambda}{2}\|\mathbf{Z} - \mathbf{W}\|^2 + \gamma\langle diag(\mathbf{W1}) - \mathbf{W}, \mathbf{S}\rangle \\ &s.t.\, \mathrm{diag}(\mathbf{W}) = 0, \mathbf{W} \ge 0, \mathbf{W} = \mathbf{W}^T,\end{aligned} \tag{21}$$

Note that the above three subproblems are convex and have closed-form solutions.

The graph-based discriminant analysis finds an ideal projection capturing essential data structure in an informative graph which is solved as a generalized eigenvalue decomposition. Compared to traditional LDA and PCA, GDA has the best discriminative ability. The low-rank property has been proven to be excellent at preserving global data structures. In the proposed GMKL algorithm, a block diagonal affine matrix is learned based on the block diagonal representation (Lu et al. 2019). Finally, the optimized combination of kernels is input to the SVM algorithm. The pseudocode of the proposed method is summarized in the following:

Algorithm 1 The proposed Algorithm (GMKL)

Input original HSI ($\mathbf{X} \in \mathbf{R}^{(I_1 \times I_2 \times I_3)}$), Ground truth ($\mathbf{G}_{th} \in \mathbf{R}^{(I_1 \times I_2)}$), SVM Parameters ($C, \alpha$), Initial and total number of superpixels I and S, Kernel scale σ, number of base kernels M, Attributes thresholds (a, Std), Training and Testing set

1: Determine the initial superpixels number $I(p)$ by Equation 1

2: Apply DBSCAN (Shen et al. 2016) segmentation;

3: for s=1, . . . ,S

Implement DBSCAN segmentation on each scale s: segMap=DBSCAN(RGB, $I(p)$);

Search ASs for target superpixel Ω_i : $\Omega_{ik} = search(segMap, \Omega_i)$, and compute the relative contributions:

$w_{ik} = \exp\left(-2r_0||\Omega_i^{mean} - \Omega_{ik}^{mean}||_2^2\right)$

Calculate base kernel $\mathbf{K}_s^{AS}$ for scale s by Equation 5

End For

4: Generate another base kernel from stacked EAPs features ($\mathbf{K}_{EAP}$)

5: Obtain **Q** matrix by vectorizing the base kernels

6: Calculate the projection matrix **P** by Equation 12 and the projection vector η_m^*

7: Create the optimum kernel $\mathbf{K}^*(\mathbf{x}_i, \mathbf{x}_j) = \sum_{m=1}^{S+1} \eta_{1m}^* \mathbf{K}_m(\mathbf{x}_i, \mathbf{x}_j)$

8: Perform SVM by setting $\mathbf{K}^*$

Output: Predicted labels

Algorithm 2 Solve (18) by Alternating Minimization

Initialize:$K = 0, \mathbf{Z}^k = 0, \mathbf{S}^k = 0, \mathbf{W}^k = 0$

while not converged to (Maximum iteration)

(1) Compute $\mathbf{S}^{k+1}$ by solving (19);

(2) Compute $\mathbf{Z}^{k+1}$ by solving (20);

(3) Compute $\mathbf{W}^{k+1}$ by solving (21);

$k = k + 1$

End while

3. Data Description

We evaluated our proposed method on two popular hyperspectral remote sensing images:

(1) Pavia University (PU): the first dataset covers Pavia University in Pavia, Italy, which was collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. It has a spatial resolution of 1.3 m per pixel and a spectral coverage ranging from 0.43 to 0.86 m, with 610 × 340 pixels and 103 spectral bands after removal of water absorption bands. As illustrated in Figure 2, this dataset contains 42,776 labelled samples grouped into 9 classes.

(2) Indian Pines (IP): Figure 3 depicts the second dataset, which was acquired by an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor with 145 × 145 pixels and 220 bands. Twenty water absorption bands are removed and 200 bands remain for experimentation. The dataset was gathered across an agricultural

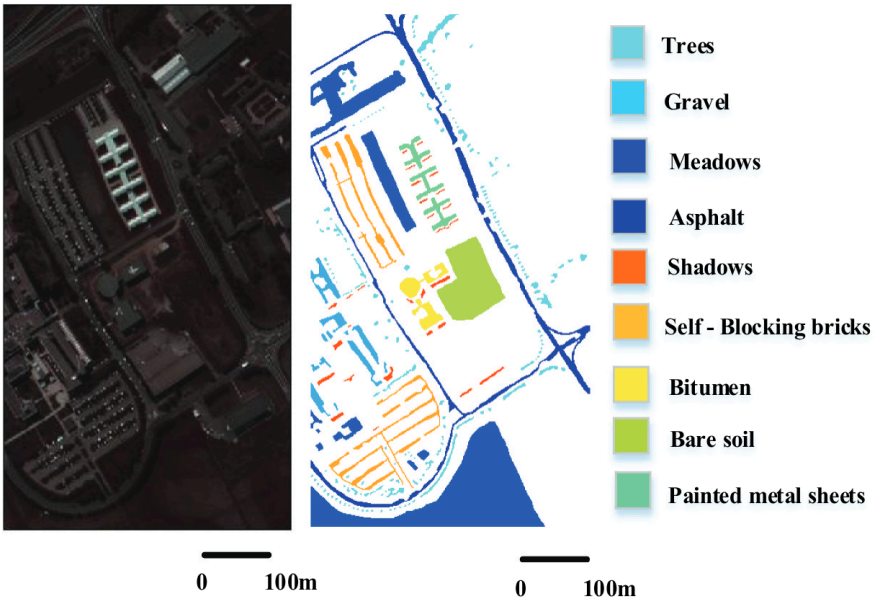


Figure 2. False-colour and ground truth map for PU dataset.

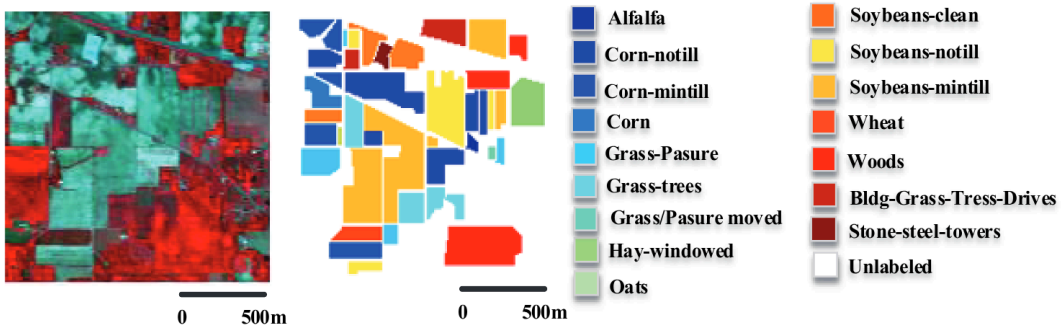


Figure 3. False-colour and ground truth map for IP dataset.

area in north-western Indiana, and it has 10,366 labelled pixels and 16 classes. This dataset has been captured with a spatial resolution of 20 m and a spectral range of 0.4 to 2.45 μm.

4. Experiments and parameter settings

As shown in Figure 1, in the proposed method different feature sets are obtained based on multiscale superpixels, and extended attributes profiles. Then, multiple Gaussian kernels are created from each feature set and given to the proposed MKL-based classifier. Typically, there are two methods to train MKL-based classifiers, one-stage methods and two-stage methods (Gu et al. 2017). We propose a two-stage method in which the base kernel weights are calculated independently of the classifier. The base kernel weights were obtained first; then, they were taken as the known conditions to solve the parameters of the classifier. To train the classifier, a 10-fold cross validation was utilized to select the optimal classifier. Each of the data was split randomly, and a (fixed) number of samples was selected without replacement as training samples and the remaining ones as test set. For each data partitioning, we retrain the classifier with the training samples and then estimate the overall accuracy (OA_i) with the test ones. The final OA is obtained as the average of the OA_i. The penalization parameter C of SVM has been optimized by cross-validation in the interval $\left[10^2, 10^3, \ldots, 10^7\right]$ for both datasets. To verify the superiority of our proposed method, different experiments were carried out. The overall accuracy (OA) and Kappa coefficient (κ) were calculated to evaluate the classification performance, and each experiment was repeated 10 times to assure the reliability of the results.

We have used a system with an Intel Core i7 (3.6 GHz) and 32 GB of RAM. Our proposed GMKL was implemented in the Google Colab environment using Keras library, and the other methods were implemented in their original environments.

In the first experiment, some conventional classification methods, such as SRC (W. Li and Du 2016), IAPs (Hong et al. 2020), MFL (J. Li et al. 2014) and MSFE (Tu et al. 2019) are referred to so that the effectiveness of using multiple kernels on classification performance can be assessed.

For SRC, the classification process is based on spectral features, and the sparse vector of each feature is computed by l_1 norm optimization with regularization parameter, like in the relevant article. In the next referred algorithm, invariant spatial and spectral features are extracted via superpixel segmentation and Fourier transformation. The output labels are then determined using a random forest classifier based on a stacked features vector. The MFL method is based on multiple linear regression (MLR) classifier and EMAPs features. The EMAPs are built with threshold values ranging from 2.5% to 10% and a step size of 2.5% for standard deviation attribute, and for area attribute, the thresholds are selected 200, 500 and 1000. For the MSFE algorithm, diverse feature sets are generated by a multi-scale Gaussian pyramid decomposition, and then single-kernel SVM is applied to each set.

In the second experiment, several outstanding MKL-based approaches such as Sparse-MKL (Gu et al. 2014), CS-MKL (T. Liu et al. 2016), DMKL (Wang, Gu, and Tuia 2016), NMKL (Gu et al. 2016) and MASEMAP-MKL (Pan et al. 2020) have been selected as benchmarks. In all these methods except the last, the first bands with 99% energy

are maintained using PCA. To extract spatial features, a diamond SE with 10 sizes, [1,2,3,4,5,6,7,8,9,10], has been utilized. Then, 10 closings and openings have been applied to each PC, 21 base kernels correspond to 10 closings, and 10 openings and PCs are created, respectively. A robust PCA is used to find the optimum kernel combination for Sparse-MKL. For the CS-MKL method, the base kernels are calculated for any two classes, and the learning process is carried out separately for each pair class. In the DMKL approach, LDA has been applied to learn the optimum kernel matrix. For NMKL, the kernel matrix in the learning model is computed based on the Hadamard product among the base kernels. Kernel learning method of MASEMAP-MKL is the same as PCA-MKL with base kernels produced from superpixels and EMAPs. Our results are also compared to another DMKL-based technique (Gu, Liu, and Li 2019), in which feature sets are extracted based on tensor decomposition. For further evaluation, the accuracy of the proposed classifier is compared to ASMGSSK (L. Sun et al. 2019), MAST-DMKL (Gu, Liu, and Li 2019), SSFTT (L. Sun et al. 2022) and Ensemble (Gu et al. 2017) at various training rates with a few samples (1%, 3% and 5%).

In the third experiment, the proposed method's performance in 50, 100 and 200 samples per class is compared to some cutting-edge NN-based methods, such as CNN (Hu et al. 2015), CNN-PPE (W. Li et al. 2017), GCNN (Hong et al. 2021), MDBRSSN (Ghaderizadeh et al. 2022), CKC (Ansari et al. 2022) and OSICN (Yang et al. 2023) to illustrate the effectiveness of GMKL.

In the proposed GMKL, three RGB bands of HSI have been separated according to their frequency using the ENVI software. For the PU image, bands 10, 27 and 46, and for the IP, bands 12, 15 and 29 are chosen as RGB. Then, [150, 250, 350, 450, 550, 650] various superpixels from the PU and [200, 300, 400, 500, 600] superpixels from the IP dataset are extracted based on the BDSCAN algorithm. Gaussian kernels with sigmas [0.5, 1] are created from each scale of superpixels, and a total of 12 and 10 feature sets, respectively, for the PU and IP images, are constructed. Two other feature sets are created by EAPs with two criteria: area and standard deviation. The area attributes are extracted with thresholds [1000, 500, 200] and threshold values ranging from 2.5% to 10% with a step size of 2.5% for the standard deviation attributes. The entire bands acquired from these features are stacked and a Gaussian kernel with variance of 1.5 corresponding to this feature set is generated (Kernel variances are selected using cross-validation).

4.1. Experimental results

The mean of overall accuracy and class-specific accuracy is reported by a standard deviation of 10 time repetitions with 1% training samples in Tables 1 and 2. Also, the computational time of testing and training procedure are computed. Only a few pixels (1%) from each class are randomly chosen as a training set, and the remaining pixels are assumed to be test samples. It is obvious that the proposed GMKL gains at least a 6.61% higher OA than the others for the PU image, and improves OA by at least 9.6% for the IP image. The results reveal that the accuracy of the proposed method is much higher than the SRC method, which only uses spectral information for classification.

Although other referenced methods utilize multiple features (spectral and spatial) to reach high accuracy, the proposed GMKL has better performance. This establishes that, in addition to considering spectral and spatial features, the optimal feature combination

Table 1. Classification results (in per cent) for PU compared with non-MKL methods.

Name of classes	Train/Test	SRC	MFL	MSFE	IAP	GMKL (proposed method)
Asphalt	66/6566	81.81 ± 0.0195	89.77 ± 0.0211	92.52 ± 0.0013	97.80 ± 0.0083	99.15 ± 0.0058
Meadows	186/18463	94.66 ± 0.0069	94.48 ± 0.0134	99.99 ± 0.0003	99.64 ± 0.0032	99.94 ± 0.0005
Gravel	21/2078	65.60 ± 0.0260	75.44 ± 0.0561	31.75 ± 0.0632	67.62 ± 0.0321	99.77 ± 0.0017
Trees	31/3033	86.63 ± 0.0280	81.52 ± 0.0258	48.82 ± 0.0565	88.33 ± 0.0292	94.80 ± 0.0180
Painted metal sheets	13/1332	99.70 ± 0.0014	98.46 ± 0.0073	99.90 ± 0.0021	99.27 ± 0.0045	99.58 ± 0.0019
Bare soil	50/4979	62.90 ± 0.0339	89.94 ± 0.0277	49.89 ± 0.1028	79.02 ± 0.0300	99.87 ± 0.0026
Bitumen	13/1317	67.87 ± 0.0776	95.08 ± 0.0240	12.17 ± 0.1601	64.88 ± 0.0718	99.26 ± 0.0104
Self-Blocking Bricks	37/3645	71.57 ± 0.0233	86.40 ± 0.0377	89.06 ± 0.0801	91.25 ± 0.0156	98.58 ± 0.0045
Shadows	9/938	88.62 ± 0.0200	57.17 ± 0.0781	25.23 ± 0.0523	96.66 ± 0.0086	99.96 ± 0.0007
OA(%)		84.14 ± 0.0066	89.97 ± 0.0083	80.60 ± 0.0161	92.67 ± 0.0054	**99.28 ± 0.0013**
$\mathcal{K}$		0.7872 ± 0.009	86.73 ± 0.0106	0.7291 ± 0.0240	0.9012 ± 0.007	**0.99 ± 0.0018**
Times(s)		1759.30	166.20	3037.2016	**61.16**	216.96

The best results are shown in boldface.

Table 2. Classification results (in per cent) for IP dataset compared with non-MKL methods.

Name of classes	Train/Test	SRC	MFL	MSFE	IAP	GMKL (proposed method)
Corn-no till	72/1362	37.88 ± 0.0641	59.65 ± 0.0627	74.84 ± 0.0432	62.62 ± 0.0622	79.47 ± 0.0887
Corn- min till	42/792	31.26 ± 0.0815	52.43 ± 0.1112	62.62 ± 0.0532	47.25 ± 0.0830	86.73 ± 0.0511
Corn	12/222	12.93 ± 0.0708	23.41 ± 0.0854	42.21 ± 0.883	19.02 ± 0.0989	63.58 ± 0.1792
Grass-pasture	25/472	64.02 ± 0.1172	76 ± 0.0808	77.49 ± 0.1445	59.40 ± 0.1536	84.49 ± 0.0774
Grass-trees	37/710	83.59 ± 0.0687	84.84 ± 0.0593	97.70 ± 0.0468	94.89 ± 0.0282	98.76 ± 0.0166
Hay-windrowed	24/465	96.32 ± 0.0327	80.93 ± 0.0759	100	96.18 ± 0.0409	99.59
Soybean-no till	48/920	49.69 ± 0.0949	63.79 ± 0.1220	72.02 ± 0.1101	56.97 ± 0.0838	82.56 ± 0.0530
Soybean-min till	123/2345	64.87 ± 0.0446	80.17 ± 0.0492	93.96 ± 0.0341	82.24 ± 0.0516	97.52 ± 0.0151
Soybean-clean	31/583	21.25 ± 0.0669	51.02 ± 0.1085	41.16 ± 0.1599	29.23 ± 0.0992	79.64 ± 0.1486
Wheat	11/201	80.24 ± 0.1431	87.76 ± 0.1376	90.94 ± 0.0734	83.58 ± 0.1495	99.84 ± 0.0033
Woods	65/1229	86.76 ± 0.0495	89.26 ± 0.0806	98.39 ± 0.0248	95.87 ± 0.0237	99.32 ± 0.0086
Bldg- Grass-Trees-Drives	19/361	20.21 ± 0.0655	52.47 ± 0.1615	51.02 ± 0.1236	43.84 ± 0.0979	92.15 ± 0.1038
OA(%)		57.32 ± 0.0156	70.83 ± 0.0562	80.83 ± 0.0188	71.06 ± 0.0194	**90.43 ± 0.0192**
$\mathcal{K}$		0.5083 ± 0.0195	0.6627 ± 0.0682	0.7766 ± 0.0223	0.6616 ± 0.0250	**0.89 ± 0.0221**
Times(s)		93.12	9.79	797.05	**8.3221**	47.05

The best results are shown in boldface.

obtained by GMKL improves the classifier's overall performance. This is mostly due to the fact that when several heterogeneous features are put together into a single feature vector, the valuable information of more representative features is diminished by other less important ones. Furthermore, the standard deviation of the proposed method is lower than the others, which confirms the most stable results.

The numerical results in Tables 3 and 4 also indicate that the proposed method outperforms popular MKL-based methods. As can be observed in the results, the GMKL algorithm attains gains ranging from 1.11% to 9.86% for PU and 1.32% to 26.13% for IP, indicating a significant improvement, particularly for the IP dataset. For a very few labelled samples, how the kernels are combined and the optimum kernel generated becomes

Table 3. Classification results (in per cent) for PU dataset compared with MKL-based methods.

Name of classes	Train/Test	NMKL (L)	SparseMKL	DMKL	CS-MKL	MASEMAP-MKL	GMKL (proposed method)
Asphalt	66/6566	95.03 ± 0.0234	94.79 ± 0.0136	94.42 ± 0.0199	96.22 ± 0.0236	97.74 ± 0.0054	99.15 ± 0.0058
Meadows	186/18463	99.16 ± 0.0042	95.64 ± 0.0104	97.26 ± 0.0068	96.08 ± 0.0242	99.86 ± 0.0004	99.94 ± 0.0005
Gravel	21/2078	76.28 ± 0.0539	74.57 ± 0.0891	84.51 ± 0.0556	80.31 ± 0.1001	89.82 ± 0.0181	99.77 ± 0.0017
Trees	31/3033	94.27 ± 0.0106	96.31 ± 0.0129	98.14 ± 0.0051	97.21 ± 0.0061	98.34 ± 0.0045	94.80 ± 0.0180
Painted metal sheets	13/1332	96.54 ± 0.0415	98.76 ± 0.0118	99.71 ± 0.0012	57.52 ± 0.4150	99.98 ± 0.0003	99.58 ± 0.0019
Bare soil	50/4979	43.14 ± 0.0772	81.48 ± 0.0400	89.55 ± 0.0410	86.22 ± 0.1012	99.15 ± 0.0043	99.87 ± 0.0026
Bitumen	13/1317	85.60 ± 0.0548	91.88 ± 0.0366	86.77 ± 0.0407	81.49 ± 0.1013	96.94 ± 0.0078	99.26 ± 0.0104
Self-Blocking Bricks	37/3645	92.79 ± 0.0328	95.08 ± 0.0142	95.01 ± 0.0302	94.53 ± 0.0246	93.33 ± 0.0078	98.58 ± 0.0045
Shadows	9/938	99.64 ± 0.0024	98.92 ± 0.0120	99.89 ± 0.0009	90.32 ± 0.2048	98.47 ± 0.0064	99.96 ± 0.0007
OA(%)		89. 42 ± 0.0107	93.34 ± 0.0083	94.97 ± 0.007	92.32 ± 0.0207	98.17 ± 0.001	**99.28 ± 0.0013**
$\mathcal{K}$		0.85 ± 0.0152	0.91 ± 0.0116	0.93 ± 0.0094	0.89 ± 0.0277	0.97 ± 0.0015	**0.99 ± 0.0018**
Times(s)		1102.60	2183.61	**64.99**	351.32	333.51	216.96

The best results are shown in boldface.

Table 4. Classification results (in per cent) for IP dataset compared with MKL-based methods.

Name of classes	Train/Test	NMKL(L)	SparseMKL	DMKL	CS-MKL	MASEMAP-MKL	GMKL (proposed method)
Corn-no till	14/1420	62.96 ± 0.0943	66.63 ± 0.0739	64.59 ± 0.1009	55.79 ± 0.2360	86.23 ± 0.0228	79.47 ± 0.0887
Corn- min till	8/826	54.98 ± 0.1034	50.54 ± 0.0863	62.68 ± 0.1036	49.39 ± 0.1596	85.19 ± 0.0209	86.73 ± 0.0511
Corn	2/232	40.82 ± 0.1791	36.64 ± 0.1302	51.77 ± 0.1899	42.46 ± 0.2550	82.43 ± 0.0552	63.58 ± 0.1792
Grass-pasture	5/492	54.15 ± 0.2770	62.01 ± 0.1552	72.85 ± 0.1425	62.93 ± 0.0225	87.35 ± 0.0453	84.49 ± 0.0774
Grass-trees	7/740	73.74 ± 0.1108	77.20 ± 0.0820	79.99 ± 0.1138	74.35 ± 0.1397	94.30 ± 0.0245	98.76 ± 0.0166
Hay-windrowed	5/484	50.08 ± 0.4790	97.44 ± 0.0355	96.82 ± 0.0231	58.57 ± 0.4282	99.20 ± 0.0048	99.59
Soybean-no till	10/958	61.88 ± 0.0676	65.18 ± 0.0770	66.10 ± 0.1053	53.77 ± 0.2029	84.71 ± 0.0355	82.56 ± 0.0530
Soybean-min till	25/2443	71.83 ± 0.0872	76.16 ± 0.0503	75.15 ± 0.0746	62.08 ± 0.2302	90.21 ± 0.0225	97.52 ± 0.0151
Soybean-clean	6/608	34.98 ± 0.2087	45.25 ± 0.0968	50.07 ± 0.1114	22.60 ± 0.1744	73.65 ± 0.0554	79.64 ± 0.1486
Wheat	2/210	77.67 ± 0.2785	94 ± 0.0547	91.24 ± 0.0769	80.62 ± 0.2771	98.66 ± 0.0055	99.84 ± 0.0033
Woods	13/1281	82.86 ± 0.2497	93.08 ± 0.0502	95.39 ± 0.0288	69.13 ± 0.2553	98.48 ± 0.0050	99.32 ± 0.0086
Bldg- Grass-Trees -Drives	4/376	51.22 ± 0.1532	55.11 ± 0.0963	60.77 ± 0.1241	46.62 ± 0.1878	83.52 ± 0.0556	92.15 ± 0.1038
OA(%)		64.30 ± 0.0513	71.04 ± 0.0216	73.38 ± 0.0299	58.01 ± 0.0846	89.11 ± 0.0096	**90.43 ± 0.0192**
$\mathcal{K}$		0.59 ± 0.0587	0.66 ± 0.0251	0.69 ± 0.0339	0.52 ± 0.9280	0.87 ± 0.0084	**0.89 ± 0.0221**
Times(s)		137.07	169.69	**21.46**	144.59	198.25	47.05

The best results are shown in boldface.

more crucial. The proposed graph-based MKL algorithm builds an efficient kernel machine to assess the similarity of observed samples and generate a more globally optimal solution for the weights of basis kernels.

In terms of computational time, subspace MKL such as DMKL and the proposed GMKL are much faster than others. It is worth noting that the computational complexity of GMKL and DMKL algorithms without considering feature extraction procedures is $O(M^3)$, whereas the complexity of other methods is $O(M \times N^3)$, where M denotes the number of base kernels and N is the number of training samples.

The results in Tables 5 and 6 tabulate the accuracies of the proposed method in comparison to some recent and popular classification methods in different low training

Table 5. Overall accuracy (in per cent) of the proposed method compared with some other methods for PU dataset in different percentages of training samples per class.

	Pavia University		
	Percentage of training samples		
Classifiers	1%	3%	5%
GMKL (Proposed)	**99.28 ± 0.0013**	**99.60 ± 0.001**	**99.71 ± 0.0008**
SSFTT	96.98 ± 0.7410	97.50 ± 0.7230	99.21 ± 0.7113
MAST-DMKL	98.41 ± 0.0312	98.95 ± 0.0321	99.25 ± 0.0302
ASMGSSK	98.46 ± 0.0491	99.47 ± 0.0465	99.69 ± 0.0394
Ensemble	96.87 ± 0.0245	98.63 ± 0.0301	99.15 ± 0.0263

Table 6. Overall accuracy (in per cent) of the proposed method compared with some other methods for IP dataset in different percentages of training samples per class.

	Indian Pines		
	Percentage of training samples		
Classifiers	1%	3%	5%
GMKL (Proposed)	**90.43 ± 0.0192**	**96.56 ± 0.0071**	**98.20 ± 0.0050**
SSFTT	82.87 ± 0.7521	92.08 ± 0.7256	94.89 ± 0.7217
MAST-DMKL	81.03 ± 0.0325	87.97 ± 0.0346	91.46 ± 0.0330
ASMGSSK	89.91 ± 0.0591	95.59 ± 0.0521	96.68 ± 0.0552
Ensemble	79.80 ± 0.0342	90.48 ± 0.0351	94.65 ± 0.0367

samples (1%, 3% and 5%). According to the results, the GMKL classifier outperforms MAST-DMKL, in which a new DMKL method based on tensor decomposition has been introduced. Additionally, the proposed GMKL improves the performance of SSFTT, which is based on a Tokenization transform and an FC layer with a softmax activation function, by at least 7.56% and 2.3% for IP and PU, respectively, in 1% training samples. In general, the accuracy of the proposed method for PU is always greater than 99%, and for Indian images, it is higher than 90%. It demonstrates the robustness of the proposed method and show how graph-based MKL can better model feature similarities.

Figures 4 and 5 illustrate the classification maps of various classifiers on the PU and IP datasets in 1% training samples. Because the SRC method does not consider spatial information, the classification map appears to be quite noisy. Other algorithms, such as MSFE, IAPs and MFL, produce a more accurate classification map with fewer misclassified samples. In comparison to the previously mentioned classification methods, GMKL achieves the best performance because it utilizes DBSCAN to take spatial information into account and classifies pixels more accurately in detailed and boundary regions. Also, the effect of the number of training samples on the compared classifiers has been illustrated in Figures 6 and 7. It is obvious that the performance of all classifiers generally progresses as the number of training samples increases. The overall accuracy of the proposed GMKL confirms its superiority. Moreover, by increasing the number of samples, the standard deviation is reduced, resulting in a more stable classifier.

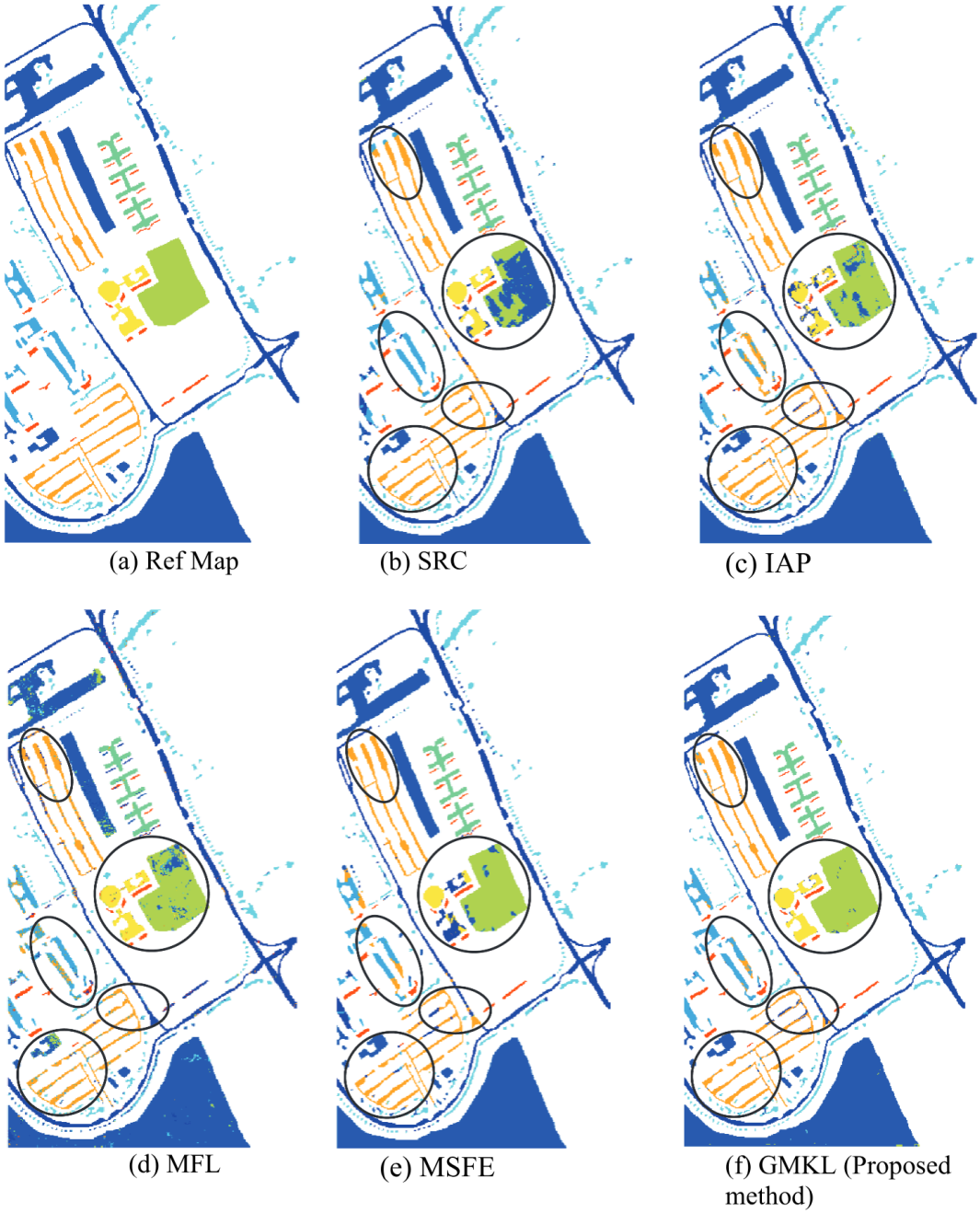


Figure 4. Classification maps of PU for different approaches in 1% training samples.

The classification map of GMKL has been compared to both ground truth and several MKL-based methods in Figures 8 and 9. It is evident that the proposed GMKL gains high discriminative capability by incorporating a multiscale structure and a graph-based MKL. As shown in Figure 8, the proposed approach better preserves the details of the PU image. Consequently, misclassified pixels in the IP dataset for Corn, Corn-min till and Soybean-no till classes are negligible when compared to other methods. It can be concluded that the proposed GMKL effectively maintains edge regions by adopting a new superpixel segmentation strategy and attains maximum class separation based on the proposed graph-based MKL.

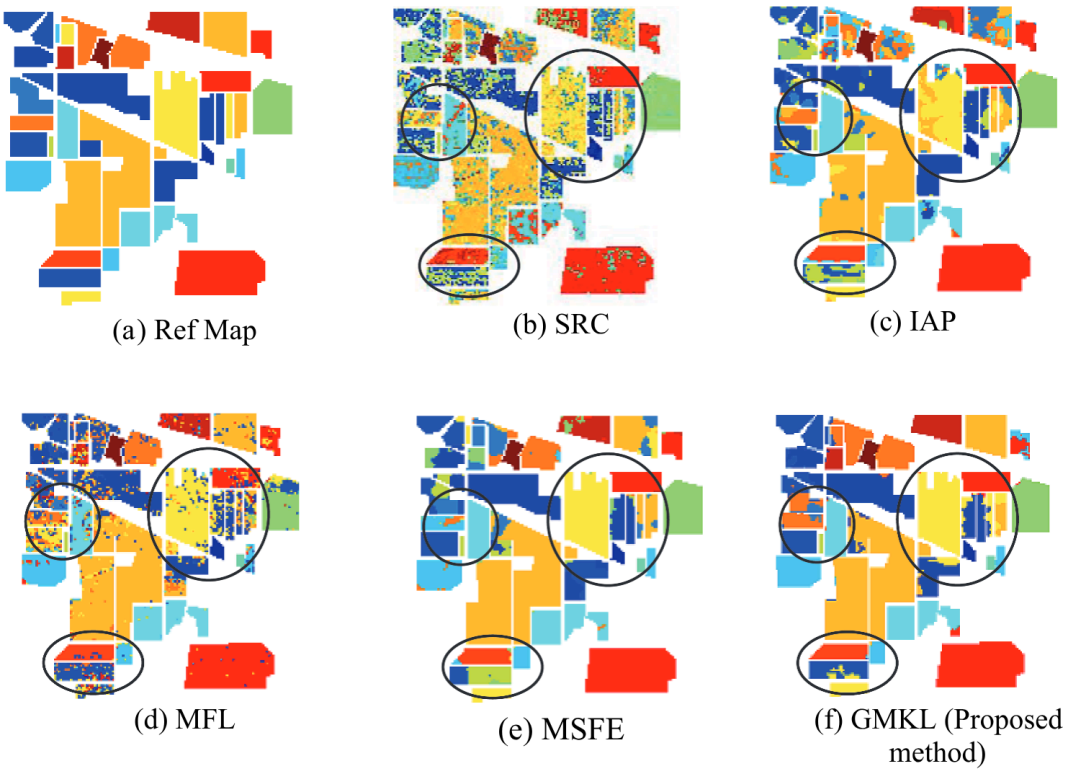


Figure 5. Classification maps of IP for different approaches in 1% training samples.

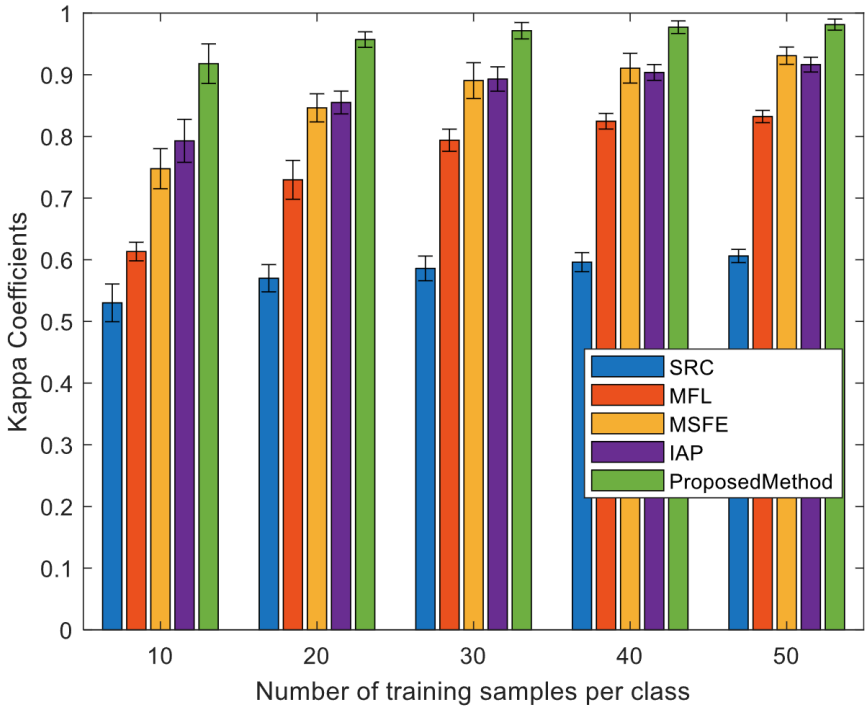


Figure 6. Kappa coefficient of PU for GMKL compared with non-MKL methods.

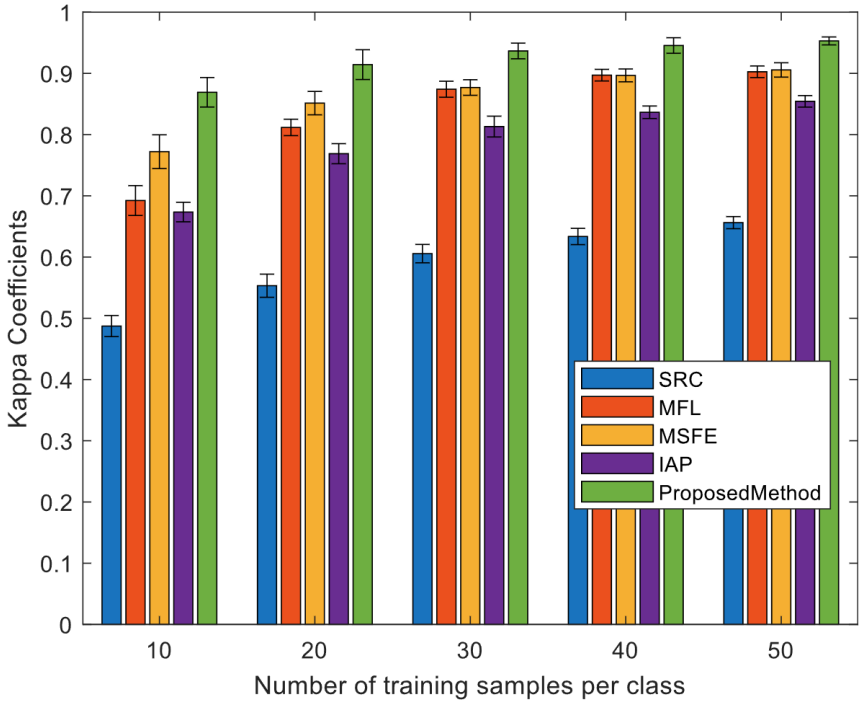


Figure 7. Kappa coefficient of IP for GMKL compared with non-MKL methods.

Figures 10 and 11 show a plot of the Kappa coefficients versus the number of training samples per class for comparing MKL-based classifiers. These results indicate that GMKL outperforms all competing methods, which validates the conclusions drawn in Tables 3 and 4. On the other hand, the standard deviation of two data sets decreases as the number of training samples increases, implying the outstanding performance of the proposed classifier. In addition, the results of several NN-based methods in Table 7 for a various numbers of training samples show that the proposed method has higher performance. This is mostly because NN-based algorithms have a large number of parameters to be learned and need more training samples to obtain a better performance.

Computational time is another important metric for assessing the efficacy of classification methods, in particular, for deep learning models. When compared to NN-based methods (Table 8), the proposed GMKL attains the maximum accuracy by effectively fusing spectral-spatial features while requiring significantly less computation time.

The effect of different number of superpixels on overall accuracy has been shown in Figures 12 and 13. As can be seen, GMKL with a low number of superpixels provides greater accuracy when the number of training samples is low. A larger number of superpixels results in higher overall accuracy, when the number of training samples is increased. This implies that small-scale superpixels cannot adequately exploit the spatial-spectral information when the number of labelled samples is limited. Furthermore, segmentation may degrade classification performance when the number of labelled samples is substantially increased.

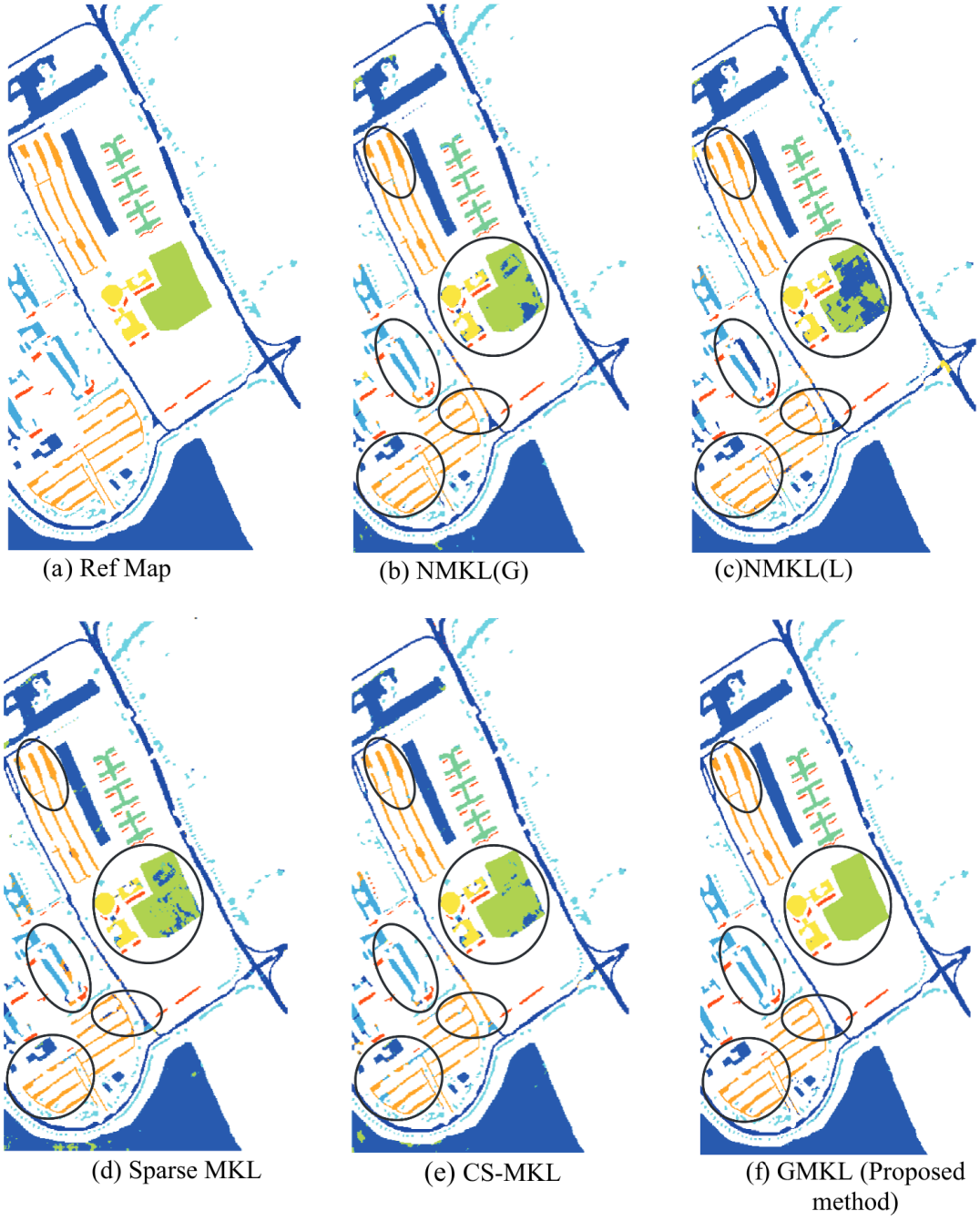


Figure 8. Classification maps of PU for the different MKL-based approaches in 1% training samples per class.

Based on the results, the proposed GMKL has superior performance to state-of-the-art methods in a small number of training samples in terms of accuracy and stability. MKL-based classifiers are feature fusion-based approaches that have much fewer parameters to be set compared to deep learning-based methods and less computational time. However, the handcrafted features used in the initial stage of our proposed method may limit the adaptation of the classification model for each data source, which leads to unsatisfactory identification and robustness.

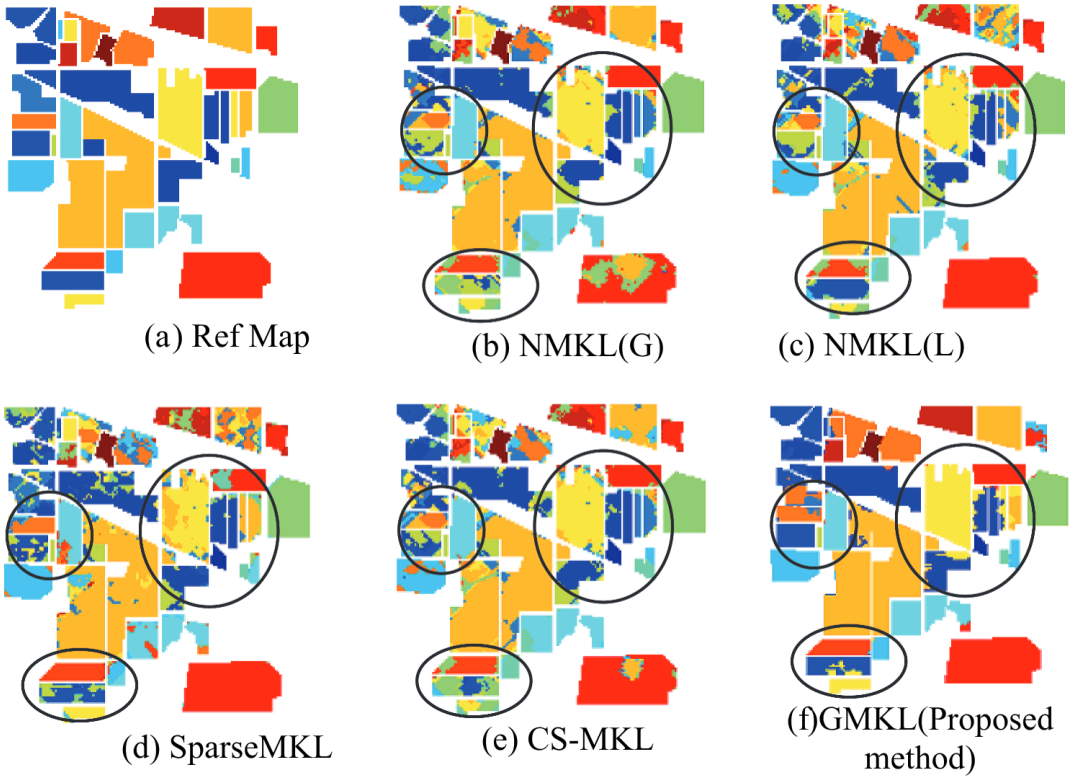


Figure 9. Classification maps of IP for the different MKL-based approaches in 1% training samples per class.

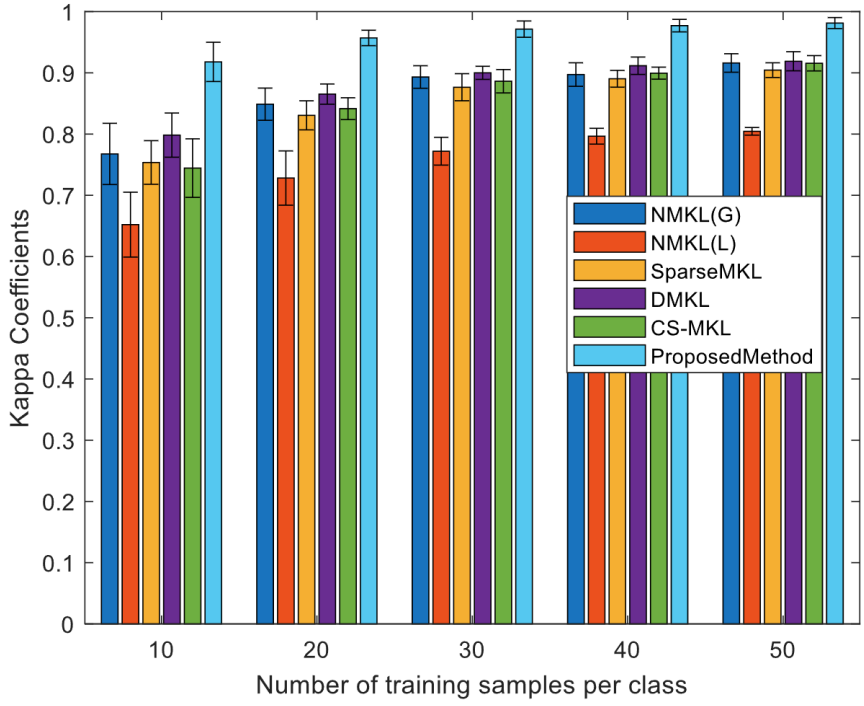


Figure 10. Kappa coefficient of PU for GMKL compared with other MKL-based methods.

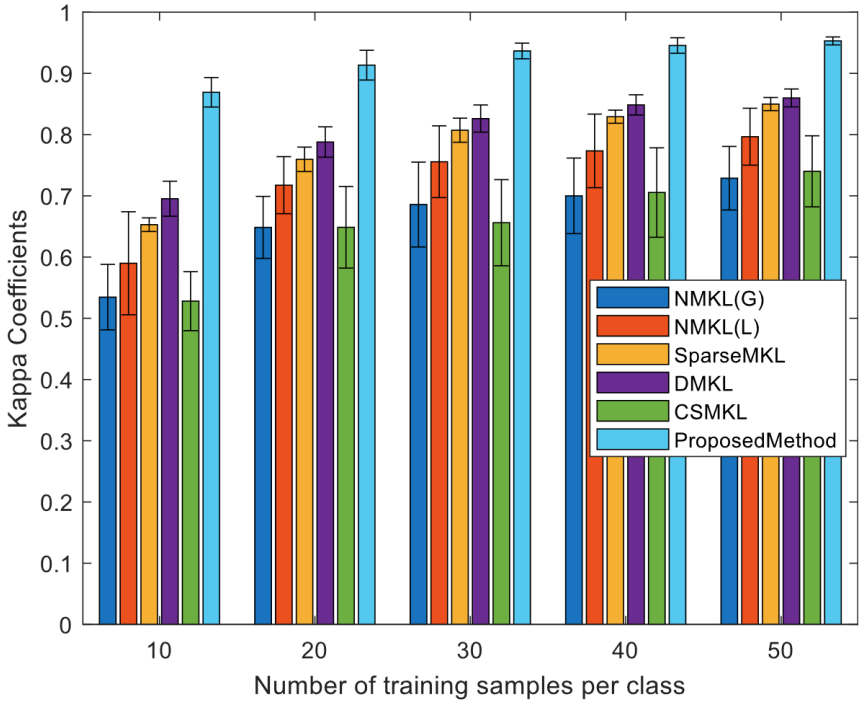


Figure 11. Kappa coefficient of IP for GMKL compared with other MKL-based methods.

Table 7. Comparison results of the proposed method in terms of overall accuracy with some state-of-the-art CNN-based methods for different numbers of training samples.

Dataset	Indian Pines						Pavia University					
Number of training Samples per class	CNN-PPE	GCN	CKC	OSICN	MDBRSSN	GMKL (proposed method)	CNN-PPE	GCN	CKC	OSICN	MDBRSSN	GMKL (proposed method)
50	88.34	79.89	88.64	90	95.83	**96.03**	88.14	83.20	90.69	94.40	98.18	**98.59**
100	91.72	84.56	92.05	92.65	96.28	**97.57**	93.35	87.67	94.76	96.73	99	**99.30**
200	93.9	92.43	94.56	94.13	97.92	**98.17**	96.45	90.20	96.54	98.11	99.27	**99.63**

Table 8. Running time (H: hour, S: second) of training and testing process for different methods.

Methods	IP	PU
CNN-PPE	1 H 180 s	1 H 421 s
GCN	1 H 331 s	1 H 506 s
CKC	201.68	287.12 s
OSICN	107.49 s	142.78 s
MDBRSSN	188.75 s	231.21 s
GMKL (proposed method)	**71.42 s**	**106.73 s**

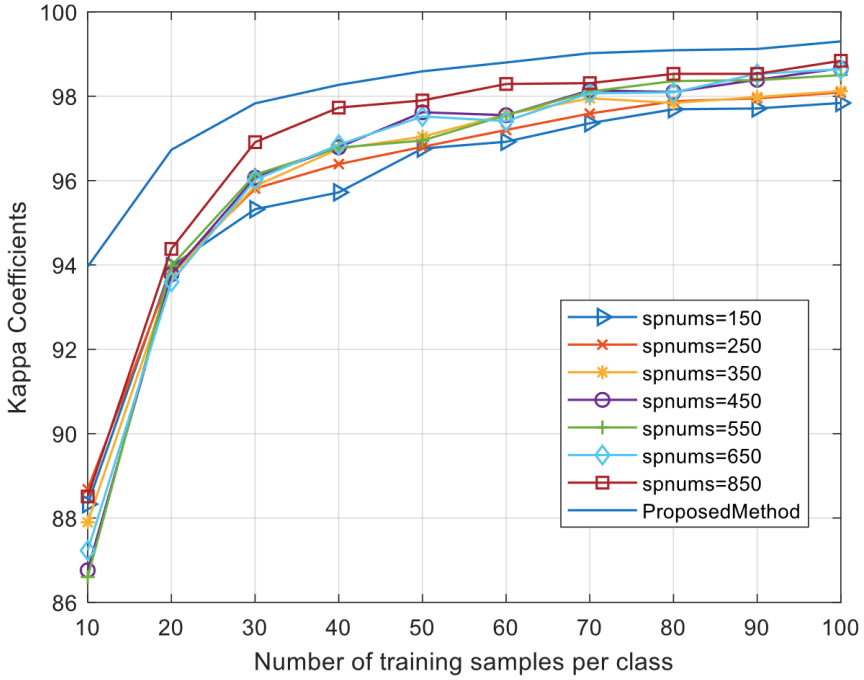


Figure 12. Classification accuracies of single-scale superpixel GMKL vs GMKL under different training samples for PU.

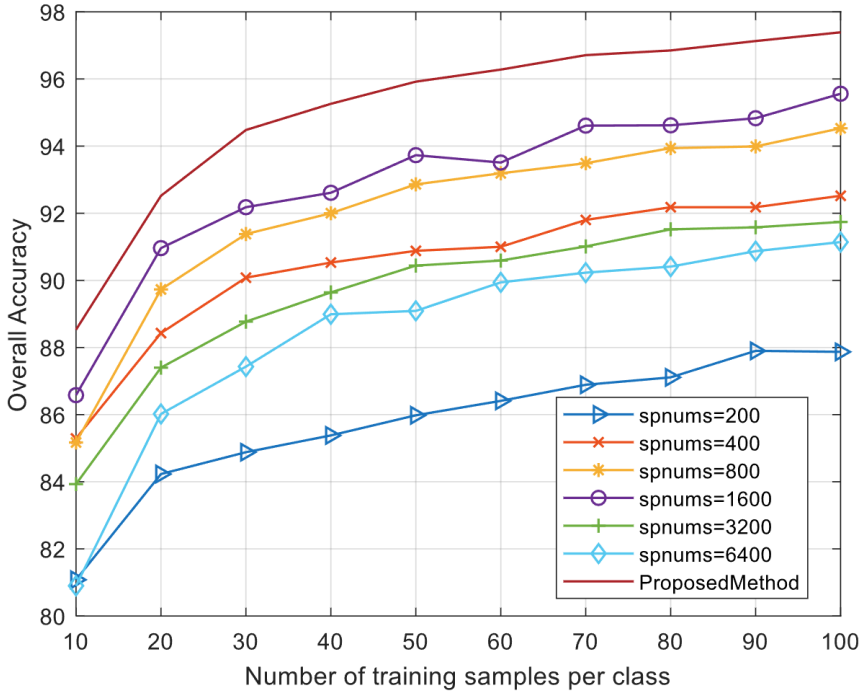


Figure 13. Classification accuracies of single-scale superpixel GMKL vs GMKL under different training samples for IP.

5. Conclusions

In this paper, a novel graph-based multiple kernel learning (GMKL) algorithm for hyperspectral image classification is proposed. In the proposed method, first, several feature sets are extracted utilizing multiscale adjacent superpixels and morphological attribute profiles. Some base kernels corresponding to each feature set are constructed. Then, an optimal combination of the kernels is learned by GMKL to maximize separability. After learning the projective direction, all basis kernels are projected to generate a discriminative kernel. Also, in the proposed method, a block diagonal representation algorithm is introduced to attain an informative block diagonal graph matrix. The computed optimum kernel has an outstanding performance in class separability, especially in the case of very low training samples such as 1%. Experimental results on popular HSI datasets confirm the superior performance of the proposed method, in terms of both quantitative metrics and visual quality in comparison with recent relevant methods.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Habibollah Danyali http://orcid.org/0000-0002-3594-1674

Data Availability statement

The dataset that supports the findings of this study is openly available to download for Pavia University dataset in Group of Inteligencia Computational (GIC) repository from [http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes] and for Indian Pines dataset from [http://lesun.weebly.com/hyperspectral-data-set.html]

References

Ansari, M., S. Homayouni, A. Safari, and S. Niazmardi. 2022. "A New Convolutional Kernel Classifier for Hyperspectral Image Classification." *IEEE Journal of Selected Topics in Applied Earth Observations & Remote Sensing* 14:11240–11256. https://doi.org/10.1109/JSTARS.2021.3123087.

Bandos, T. V., L. Bruzzone, and G. Camps-Valls. 2009. "Classification of Hyperspectral Images with Regularized Linear Discriminant Analysis." *IEEE Transactions on Geoscience & Remote Sensing* 47 (3): 862–873. https://doi.org/10.1109/tgrs.2008.2005729.

Bioucas-Dias, J. M., A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot. 2013. "Hyperspectral Remote Sensing Data Analysis and Future Challenges." *IEEE Geoscience and remote sensing magazine* 1 (2): 6–36. https://doi.org/10.1109/MGRS.2013.2244672.

Bo, C., H. Lu, and D. Wang. 2017. "Weighted Generalized Nearest Neighbor for Hyperspectral Image Classification." *Institute of Electrical and Electronics Engineers Access* 5:1496–1509. https://doi.org/10.1109/ACCESS.2017.2669149.

Briottet, X., Y. Boucher, A. Dimmeler, A. Malaplate, A. Cini, M. Diani, H. H. P. T. Bekman, P. Schwering, T. Skauli, and I. Kasen. 2006. "Military Applications of Hyperspectral Imagery." *Targets and Backgrounds XII: Characterization and Representation, SPIE* 6239:82–89. https://doi.org/10.1117/12.672030.

Cao, X., R. Li, Y. Ge, B. Wu, and L. Jiao. 2019. "Densely Connected Deep Random Forest for Hyperspectral Imagery Classification." *International Journal of Remote Sensing* 40 (9): 3606–3622. https://doi.org/10.1080/01431161.2018.1547932.

Deng, Y.-J., H.-C. Li, K. Fu, Q. Du, and W. J. Emery. 2018. "Tensor Low-Rank Discriminant Embedding for Hyperspectral Image Dimensionality Reduction." *IEEE Transactions on Geoscience & Remote Sensing* 56 (12): 7183–7194. https://doi.org/10.1109/TGRS.2018.2849085.

Fang, L., N. He, S. Li, P. Ghamisi, and J. Atli Benediktsson. 2018. "Extinction Profiles Fusion for Hyperspectral Images Classification." *IEEE Transactions on Geoscience & Remote Sensing* 56 (3): 1803–1815. https://doi.org/10.1109/tgrs.2017.2768479.

Ghaderizadeh, S., D. Abbasi-Moghadam, A. Sharifi, A. Tariq, and S. Qin. 2022. "Multiscale Dual-Branch Residual Spectral–Spatial Network with Attention for Hyperspectral Image Classification." *IEEE Journal of Selected Topics in Applied Earth Observations & Remote Sensing* 15:5455–5467. https://doi.org/10.1109/jstars.2022.3188732.

Ghojogh, B., F. Karray, and M. Crowley. 2019. "Eigenvalue and Generalized Eigenvalue Problems: Tutorial." *arXiv Preprint arXiv: 190311240*. https://doi.org/10.48550/arXiv.1903.11240.

Götze, C., F. Beyer, and C. Gläßer. 2016. "Pioneer Vegetation As an Indicator of the Geochemical Parameters in Abandoned Mine Sites Using Hyperspectral Airborne Data." *Environmental Earth Sciences* 75 (7): 1–14. https://doi.org/10.1007/s12665-016-5367-1.

Gu, Y., J. Chanussot, X. Jia, and J. Atli Benediktsson. 2017. "Multiple Kernel Learning for Hyperspectral Image Classification: A Review." *IEEE Transactions on Geoscience & Remote Sensing* 55 (11): 6547–6565. https://doi.org/10.1109/tgrs.2017.2729882.

Gu, Y., G. Gao, D. Zuo, and D. You. 2014. "Model Selection and Classification with Multiple Kernel Learning for Hyperspectral Images via Sparsity." *IEEE Journal of Selected Topics in Applied Earth Observations & Remote Sensing* 7 (6): 2119–2130. https://doi.org/10.1109/JSTARS.2014.2318181.

Gu, Y., T. Liu, X. Jia, J. Atli Benediktsson, and J. Chanussot. 2016. "Nonlinear Multiple Kernel Learning with Multiple-Structure-Element Extended Morphological Profiles for Hyperspectral Image Classification." *IEEE Transactions on Geoscience & Remote Sensing* 54 (6): 3235–3247. https://doi.org/10.1109/tgrs.2015.2514161.

Gu, Y., T. Liu, and J. Li. 2019. "Superpixel Tensor Model for Spatial–Spectral Classification of Remote Sensing Images." *IEEE Transactions on Geoscience & Remote Sensing* 57 (7): 4705–4719. https://doi.org/10.1109/TGRS.2019.2892516.

Gu, Y., C. Wang, D. You, Y. Zhang, S. Wang, and Y. Zhang. 2012. "Representative Multiple Kernel Learning for Classification in Hyperspectral Imagery." *IEEE Transactions on Geoscience & Remote Sensing* 50 (7): 2852–2865. https://doi.org/10.1109/tgrs.2011.2176341.

Hang, R., Q. Liu, D. Hong, and P. Ghamisi. 2019. "Cascaded Recurrent Neural Networks for Hyperspectral Image Classification." *IEEE Transactions on Geoscience & Remote Sensing* 57 (8): 5384–5394. https://doi.org/10.1109/TGRS.2019.2899129.

Hassanzadeh, S., H. Danyali, and M. Sadegh Helfroush. 2022a. "Combined Spatial-Spectral Schroedinger Eigenmaps with Multiple Kernel Learning for Hyperspectral Image Classification Using a Low Number of Training Samples." *Canadian Journal of Remote Sensing* 48 (5): 1–13. https://doi.org/10.1080/07038992.2021.1978840.

Hassanzadeh, S., H. Danyali, and M. Sadegh Helfroush. 2022b. "Hyperspectral Images Classification Based on Multiple Kernel Learning Using SWT and KMNF with Few Training Samples." *Journal of Spatial Science* 68 (4): 593–613. https://doi.org/10.1080/14498596.2022.2097962.

He, K., W. Sun, G. Yang, X. Meng, K. Ren, J. Peng, and Q. Du. 2022. "A Dual Global–Local Attention Network for Hyperspectral Band Selection." *IEEE Transactions on Geoscience & Remote Sensing* 60:1–13. https://doi.org/10.1109/TGRS.2022.3169018.

Hong, D., L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot. 2021. "Graph Convolutional Networks for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing*. https://doi.org/10.1109/TGRS.2020.3015157.

Hong, D., X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. Xiang Zhu. 2020. "Invariant Attribute Profiles: A Spatial-Frequency Joint Feature Extractor for Hyperspectral Image Classification." *IEEE Transactions on Geoscience & Remote Sensing* 58 (6): 3791–3808. https://doi.org/10.1109/TGRS.2019.2957251.

Honkavaara, E., T. Hakala, L. Markelin, A. Jaakkola, H. Saari, H. Ojanen, I. Pölönen, S. Tuominen, R. Näsi, and T. Rosnell. 2014. "Autonomous Hyperspectral UAS Photogrammetry for Environmental Monitoring Applications." *ISPRS Archives*. https://doi.org/10.5194/isprsarchives-XL-1-155-2014.

Hou, T., W. Sun, C. Chen, G. Yang, X. Meng, and J. Peng. 2022. "Marine Floating Raft Aquaculture Extraction of Hyperspectral Remote Sensing Images Based Decision Tree Algorithm." *International Journal of Applied Earth Observation and Geoinformation* 111:102846. https://doi.org/10.1016/j.jag.2022.102846.

Huang, Y., J. Peng, N. Chen, W. Sun, Q. Du, K. Ren, and K. Huang. 2023. "Cross-Scene Wetland Mapping on Hyperspectral Remote Sensing Images Using Adversarial Domain Adaptation Network." *Isprs Journal of Photogrammetry & Remote Sensing* 203:37–54. https://doi.org/10.1016/j.isprsjprs.2023.07.009.

Huang, Y., J. Peng, W. Sun, N. Chen, Q. Du, Y. Ning, and H. Su. 2022. "Two-Branch Attention Adversarial Domain Adaptation Network for Hyperspectral Image Classification." *IEEE Transactions on Geoscience & Remote Sensing* 60:1–13. https://doi.org/10.1109/TGRS.2022.3215677.

Hu, W., Y. Huang, L. Wei, F. Zhang, and H. Li. 2015. "Deep Convolutional Neural Networks for Hyperspectral Image Classification." *Journal of Sensors* 2015. https://doi.org/10.1155/2015/258619.

Jiang, J., J. Ma, C. Chen, Z. Wang, Z. Cai, and L. Wang. 2018. "SuperPCA: A Superpixelwise PCA Approach for Unsupervised Feature Extraction of Hyperspectral Imagery." *IEEE Transactions on Geoscience & Remote Sensing* 56 (8): 4581–4593. https://doi.org/10.1109/tgrs.2018.2828029.

Licciardi, G., P. Reddy Marpu, J. Chanussot, and J. Atli Benediktsson. 2012. "Linear versus Nonlinear PCA for the Classification of Hyperspectral Data Based on the Extended Morphological Profiles." *IEEE Geoscience & Remote Sensing Letters* 9 (3): 447–451. https://doi.org/10.1109/LGRS.2011.2172185.

Li, W., and Q. Du. 2016. "A Survey on Representation-Based Classification and Detection in Hyperspectral Remote Sensing Imagery." *Pattern Recognition Letters* 83:115–123. https://doi.org/10.1016/j.patrec.2015.09.010.

Li, W., F. Feng, H. Li, and Q. Du. 2018. "Discriminant Analysis-Based Dimension Reduction for Hyperspectral Image Classification: A Survey of the Most Recent Advances and an Experimental Comparison of Different Techniques." *IEEE Geoscience and Remote Sensing Magazine* 6 (1): 15–34. https://doi.org/10.1109/MGRS.2018.2793873.

Li, J., X. Huang, P. Gamba, J. M. Bioucas-Dias, L. Zhang, J. Atli Benediktsson, and A. Plaza. 2014. "Multiple Feature Learning for Hyperspectral Image Classification." *IEEE Transactions on Geoscience & Remote Sensing* 53 (3): 1592–1606. https://doi.org/10.1109/TGRS.2014.2345739.

Li, H.-C., Z.-X. Lin, T.-Y. Ma, X.-L. Zhao, A. Plaza, and W. J. Emery. 2023. "Hybrid Fully-Connected Tensorized Compression Network for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing*. https://doi.org/10.1109/TGRS.2023.3241193.

Li, W., J. Liu, and Q. Du. 2016. "Sparse and Low-Rank Graph for Discriminant Analysis of Hyperspectral Imagery." *IEEE Transactions on Geoscience & Remote Sensing* 54 (7): 4094–4105. https://doi.org/10.1109/tgrs.2016.2536685.

Liu, T., Y. Gu, X. Jia, J. Atli Benediktsson, and J. Chanussot. 2016. "Class-Specific Sparse Multiple Kernel Learning for Spectral–Spatial Hyperspectral Image Classification." *IEEE Transactions on Geoscience & Remote Sensing* 54 (12): 7351–7365. https://doi.org/10.1109/TGRS.2016.2600522.

Liu, W., K. Liu, W. Sun, G. Yang, K. Ren, X. Meng, and J. Peng. 2023. "Self-Supervised Feature Learning Based on Spectral Masking for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing*. https://doi.org/10.1109/TGRS.2023.3310489.

Liu, Q., J. Peng, N. Chen, W. Sun, Y. Ning, and Q. Du. 2023. "Category-Specific Prototype Self-Refinement Contrastive Learning for Few-Shot Hyperspectral Image Classification." *IEEE Transactions on Geoscience & Remote Sensing* 61:1–16. https://doi.org/10.1109/TGRS.2023.3317077.

Liu, Y., H. Pu, and D.-W. Sun. 2017. "Hyperspectral Imaging Technique for Evaluating Food Quality and Safety During Various Processes: A Review of Recent Applications." *Trends in Food Science & Technology* 69:25–35. https://doi.org/10.1016/j.tifs.2017.08.013.

Li, W., G. Wu, F. Zhang, and Q. Du. 2017. "Hyperspectral Image Classification Using Deep Pixel-Pair Features." *IEEE Transactions on Geoscience & Remote Sensing* 55 (2): 844–853. https://doi.org/10.1109/TGRS.2016.2616355.

Lu, C., J. Feng, Z. Lin, T. Mei, and S. Yan. 2019. "Subspace Clustering by Block Diagonal Representation." *IEEE Transactions Pattern Analysis Machine Intelligent* 41 (2): 487–501. https://doi.org/10.1109/TPAMI.2018.2794348.

Lu, C., J. Feng, S. Yan, and Z. Lin. 2018. "A Unified Alternating Direction Method of Multipliers by Majorization Minimization." *IEEE Transactions Pattern Analysis Machine Intelligent* 40 (3): 527–541. https://doi.org/10.1109/TPAMI.2017.2689021.

Ma, X., H. Wang, and J. Geng. 2016. "Spectral–Spatial Classification of Hyperspectral Image Based on Deep Auto-Encoder." *IEEE Journal of Selected Topics in Applied Earth Observations & Remote Sensing* 9 (9): 4073–4085. https://doi.org/10.1109/JSTARS.2016.2517204.

Mura, D., J. A. B. Mauro, B. Waske, and L. Bruzzone. 2010. "Morphological Attribute Profiles for the Analysis of Very High Resolution Images." *IEEE Transactions on Geoscience & Remote Sensing* 48 (10): 3747–3762. https://doi.org/10.1109/TGRS.2010.2048116.

Pan, L., C. He, Y. Xiang, and L. Sun. 2020. "Multiscale Adjacent Superpixel-Based Extended Multi-Attribute Profiles Embedded Multiple Kernel Learning Method for Hyperspectral Classification." *Remote Sensing* 13 (1): 50. https://doi.org/10.3390/rs13010050.

Shen, J., X. Hao, Z. Liang, Y. Liu, W. Wang, and L. Shao. 2016. "Real-Time Superpixel Segmentation by DBSCAN Clustering Algorithm." *IEEE Transactions on Image Processing* 25 (12): 5933–5942. https://doi.org/10.1109/TIP.2016.2616302.

Sheykhmousa, M., M. Mahdianpari, H. Ghanbari, F. Mohammadimanesh, P. Ghamisi, and S. Homayouni. 2020. "Support Vector Machine versus Random Forest for Remote Sensing Image Classification: A Meta-Analysis and Systematic Review." *IEEE Journal of Selected Topics in Applied Earth Observations & Remote Sensing* 13:6308–6325. https://doi.org/10.1109/jstars.2020.3026724.

Sun, Y., B. Liu, X. Yu, A. Yu, P. Zhang, and Z. Xue. 2023. "Exploiting Discriminative Advantage of Spectrum for Hyperspectral Image Classification: SpectralFormer Enhanced by Spectrum Motion Feature." *IEEE Geoscience & Remote Sensing Letters* 20:1–5. https://doi.org/10.1109/LGRS.2022.3228531.

Sun, L., C. Ma, Y. Chen, H. Jae Shim, Z. Wu, and B. Jeon. 2019. "Adjacent Superpixel-Based Multiscale Spatial-Spectral Kernel for Hyperspectral Classification." *IEEE Journal of Selected Topics in Applied Earth Observations & Remote Sensing* 12 (6): 1905–1919. https://doi.org/10.1109/JSTARS.2019.2915588.

Sun, L., G. Zhao, Y. Zheng, and Z. Wu. 2022. "Spectral–Spatial Feature Tokenization Transformer for Hyperspectral Image Classification." *IEEE Transactions on Geoscience & Remote Sensing* 60:1–14. https://doi.org/10.1109/TGRS.2022.3144158.

Sven, S., R. J. Murphy, and A. Melkumyan. 2014. "Evaluating the Performance of a New Classifier – the GP-OAD: A Comparison with Existing Methods for Classifying Rock Type and Mineralogy from Hyperspectral Imagery." *Isprs Journal of Photogrammetry & Remote Sensing* 98:145–156. https://doi.org/10.1016/j.isprsjprs.2014.09.016.

Tu, B., N. Li, L. Fang, D. He, and P. Ghamisi. 2019. "Hyperspectral Image Classification with Multi-Scale Feature Extraction." *Remote Sensing* 11 (5): 534. https://doi.org/10.3390/rs11050534.

Tu, B., C. Zhou, D. He, S. Huang, and A. Plaza. 2020. "Hyperspectral Classification with Noisy Label Detection via Superpixel-To-Pixel Weighting Distance." *IEEE Transactions on Geoscience & Remote Sensing* 58 (6): 4116–4131. https://doi.org/10.1109/tgrs.2019.2961141.

Wang, Q., Y. Gu, and D. Tuia. 2016. "Discriminative Multiple Kernel Learning for Hyperspectral Image Classification." *IEEE Transactions on Geoscience & Remote Sensing* 54 (7): 3912–3927. https://doi.org/10.1109/TGRS.2016.2530807.

Xing, F., H. Yao, Y. Liu, X. Dai, R. L. Brown, and D. Bhatnagar. 2019. "Recent Developments and Applications of Hyperspectral Imaging for Rapid Detection of Mycotoxins and Mycotoxigenic Fungi in Food Products." *Critical Reviews in Food Science and Nutrition* 59 (1): 173–180. https://doi.org/10.1080/10408398.2017.1363709.

Yang, J., D. Bo, and Z. Liangpei. 2023. "Overcoming the Barrier of Incompleteness: A Hyperspectral Image Classification Full Model." *IEEE transactions on neural networks and learning systems*. https://doi.org/10.1109/TNNLS.2023.3279377.

Yang, J., B. Du, Y. Xu, and L. Zhang. 2023. "Can Spectral Information Work While Extracting Spatial Distribution?—An Online Spectral Information Compensation Network for HSI Classification." *IEEE transactions on image processing*. https://doi.org/10.1109/TIP.2023.3244414.

Yang, J., B. Du, and L. Zhang. 2023. "From Center to Surrounding: An Interactive Learning Framework for Hyperspectral Image Classification." *Isprs Journal of Photogrammetry & Remote Sensing* 197:145–166. https://doi.org/10.1016/j.isprsjprs.2023.01.024.

Yang, G., K. Huang, W. Sun, X. Meng, D. Mao, and Y. Ge. 2022. "Enhanced Mangrove Vegetation Index Based on Hyperspectral Images for Mapping Mangrove." *Isprs Journal of Photogrammetry & Remote Sensing* 189:236–254. https://doi.org/10.1016/j.isprsjprs.2022.05.003.

Yang, J., C. Wu, B. Du, and L. Zhang. 2021. "Enhanced Multiscale Feature Fusion Network for HSI Classification." *IEEE Transactions on Geoscience and Remote Sensing* 59 (12): 10328–10347. https://doi.org/10.1109/TGRS.2020.3046757.

Zhang, M., W. Li, X. Zhao, H. Liu, R. Tao, and Q. Du. 2023. "Morphological Transformation and Spatial-Logical Aggregation for Tree Species Classification Using Hyperspectral Imagery." *IEEE Transactions on Geoscience & Remote Sensing* 61:1–12. https://doi.org/10.1109/TGRS.2022.3233847.

Zhang, L., S. Lu, C. Hu, D. Xiang, T. Liu, and Y. Su. 2021. "Superpixel Generation for SAR Imagery Based on Fast DBSCAN Clustering with Edge Penalty." *IEEE Journal of Selected Topics in Applied Earth Observations & Remote Sensing* 15:804–819. https://doi.org/10.1109/JSTARS.2021.3131187.

Zhang, L., and F. Luo. 2020. "Review on Graph Learning for Dimensionality Reduction of Hyperspectral Image." *Geo-Spatial Information Science* 23 (1): 98–106. https://doi.org/10.1080/10095020.2020.1720529.